

**TOWARDS NON-INVASIVE, EVERYDAY HEALTH DIAGNOSTICS
USING INTELLIGENT SENSING TECHNIQUES**

by

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Declaration

I hereby declare that this project report is my original work and it
has been written by me in its entirety.

I have duly acknowledged all the sources of information which have
been used in this report.

This report has also not been submitted for any
degree in any university previously.

A handwritten signature in black ink, appearing to read 'nel', with a long horizontal stroke extending to the left.

Toh Yun Sheng Nelson

10 April 2026

To my friends and family

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I am grateful for the project supervision by Prof. Wang Jingxian and Prof. Cheng Penghui. Their inputs and advice ensured that I stayed on track for the project, without which I would have found very challenging.

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Abstract

Chronic Kidney Disease (CKD) is a serious disease that motivates interest in sensing approaches that could support earlier and more accessible monitoring outside conventional laboratory workflows. As an initial feasibility study toward this broader goal, this project investigates whether Ultra-High Frequency (UHF) radio sensing can detect concentration-dependent differences in aqueous gold and silver nanoparticle solutions, viewed as candidate contrast-agent systems, by measuring transmitted and reflected radio waves. The results show that measurable concentration-dependent RF signatures can be extracted under controlled laboratory conditions, although the trends were not consistently detected across all settings. Nevertheless, the measured data supported multiclass classification across the different experimental setups, suggesting tentatively that future studies could enhance the experimental parameters and setup, and further improve the prospects of RF sensing as a viable and accessible diagnostic tool for CKD.

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Chapter 1

Introduction

Among other bodily functions, the human kidney acts to filter unwanted materials from the blood, expelling them via urination. In kidney dysfunction, the kidney no longer filters the unwanted materials as readily, resulting in a buildup of waste products in the bloodstream. Long-term and progressive dysfunction of the kidneys is referred to as Chronic Kidney Disease (CKD).

Globally, it is estimated that the age-adjusted incidence of CKD exceeds 10% and is a leading cause of deaths. In addition, it is a major contributor to disability-adjusted life years (DALYs) which measures the years lived in disability. A high DALY imposes a large disease burden on the population, beyond immediate impacts to individuals, due to loss of economic productivity and cost of caregiving.[1]

There is strong motivation therefore, to support early detection of CKD for timely interventions medically or in lifestyle changes to prevent further disease progression. Currently among other methods, CKD is commonly diagnosed and staged based on the kidneys' inferred filtration performance, for example by measuring the urine albumin-creatinine ratio (uACR) over a period of 3 months.[2]

However, early kidney damage may not be detected by such means because compensatory hyperfiltration in the remaining nephrons can preserve overall filtration despite nephron loss, and some forms of CKD may progress through nonalbuminuric pathways. Therefore, while uACR is a recommended and often early marker of CKD, uACR can miss a subset of early disease. Also, albumin and creatinine being part of normal physiology, means that the albumin-creatinine ratio is biologically variable with short-run conditions such as exercising and fever, necessitating a longer monitoring period for observing persistent abnormalities.[3]

Similar alternative non-invasive diagnostic means are being explored, such as

in the use of ingested contrast agents engineered for detection by specific means. These potentially offer complementary solutions with the existing suite of diagnostic tools and can contribute to improvements for diagnosing and staging CKD. As an initial exploratory step, this study focuses specifically on aqueous gold and silver nanoparticle solutions as candidate contrast-agent systems.

1.1 Problem Statement

1.1.1 Need for Alternative Sensing Modality

Existing kidney-related monitoring approaches rely primarily on conventional biochemical testing and are not inherently suited for convenient, repeated, everyday sensing. This creates a need to explore alternative sensing modalities that may support more accessible monitoring workflows.

1.1.2 Uncertainty in Electromagnetic Detectability

The use of gold and silver nanoparticle samples as contrast agents was explored, using the supposed difference in the samples' response to electromagnetic radiation. However, it remained unclear whether these effects can be detected consistently under practical laboratory conditions.

1.1.3 Lack of Robust Measurement Methodology

The weak nature of the expected signal differences makes it difficult to distinguish true sample-dependent effects from measurement artefacts. A robust and reproducible methodology is therefore needed for acquiring, correcting, and analysing measurements of the nanoparticle samples' response to electromagnetic radiation.

1.1.4 Finding Meaningful Correlations

Even after measurements across the different nanoparticle types and concentrations are acquired, it is not straightforward to determine whether the observed electromagnetic responses are statistically meaningful, reproducible, and sufficiently discriminative for downstream classification or predictive modelling. A structured

analytical framework is needed to evaluate feature relevance, group separability, and modelling potential.

1.2 Summary of Contributions

- Designed the experimental physical setup and overall measurement framework for studying the electromagnetic response of water-based gold and silver nanoparticle samples, motivated by the broader goal of exploring non-invasive sensing approaches for concentration-dependent contrast-agent behaviour.
- Conducted UHF simulation studies to identify a suitable operating region for physical testing and to establish expected concentration-dependent response behaviour in both transmission and reflection measurements.
- Developed a multi-stage measurement and signal-processing methodology to improve the robustness of weak RF (Radio Frequency) measurements, including offset-frequency demodulation, carrier frequency offset correction, phasor aggregation, and standard sample bracketing.
- Experimentally demonstrated that measurable RF magnitude and phase differences could be extracted across the different nanoparticle types and concentration levels under controlled conditions, although the observed relationships were not uniformly monotonic across all settings.
- Applied statistical analysis and machine learning classification to evaluate the reproducibility, separability, and downstream analytical value of the extracted RF features across the four experimental settings.

Chapter 2

Literature Review

2.1 Alternative Sensing Approaches

Current CKD evaluation relies primarily on established biochemical markers such as urine albumin measurement and assessment of glomerular filtration rate. However, such approaches are fundamentally laboratory-based and episodic, rather than naturally suited to convenient, repeated, everyday sensing. [2, 3]

This motivates interest in complementary sensing approaches that support earlier or more accessible monitoring. Non-invasive sensing strategies have attracted increasing attention in nephrology, including approaches based on engineered probes or contrast agents designed to produce detectable physical signatures. [4]

In particular, Yu et al. [5] demonstrated that near-infrared-emitting gold nanoparticles can be used for non-invasive in vivo detection of kidney dysfunction, with fluorescence imaging further enabling differentiation between stages of dysfunction. Despite its promise as an effective non-invasive imaging approach, the method reported by Yu et al. [5] still depends on specialised fluorescence imaging equipment, and is therefore less readily aligned with convenient, repeated, everyday monitoring by end users.

One possible route towards more accessible sensing is to investigate modalities that may eventually be better aligned with ubiquitous consumer devices such as smartphones, which are already widely distributed and contain integrated radio hardware for emitting and receiving electromagnetic signals. This motivates the question of whether RF-based sensing approaches could, in the longer term, provide a more practical basis for convenient and repeated monitoring.

At a broad hardware level, such a direction is plausible, since modern smart-

phones already integrate multiple antenna subsystems spanning a wide range of operating bands, including low-band and mid/high-band cellular ranges such as 698–960 MHz and 1710–2690 MHz, as well as Wi-Fi/Bluetooth bands at 2.4–2.484 GHz and 5.15–5.875 GHz.[6].

However, consumer smartphones are not presently exposed to application developers as general-purpose RF measurement. Instead, operating systems typically provide controlled, high-level interfaces for communication and proximity-related tasks. In addition, it is unclear whether their antenna designs are intrinsically suitable for the present non-invasive sensing purpose. This is because modern smartphone antennas are primarily developed under communication-driven design objectives and constraints, rather than sensing-driven ones.[6]

Nonetheless, RF sensing represents a plausible direction for further exploration, given that hardware capable of transmitting and receiving electromagnetic signals is already widely deployed in mass-market consumer devices such as smartphones.

2.2 Aqueous Nanoparticle Contrast Agent

Following a similar approach to Yu et al. [5], but exploring the use of RF sensing instead, this section discusses the use and compatibility of aqueous nanoparticle solutions as contrast agents.

At radio and microwave frequencies, the electromagnetic response of a material is commonly described in terms of its effective complex permittivity. Under sinusoidal excitation, this may be written as:

$$\varepsilon_{\text{eff}}^*(\omega) = \varepsilon'(\omega) - j \left(\varepsilon''(\omega) + \frac{\sigma}{\omega} \right) \quad (2.1)$$

where $\varepsilon'(\omega)$ and $\varepsilon''(\omega)$ are respectively the real and imaginary parts of the complex permittivity, ω is the angular frequency, and σ is the electrical conductivity. Physically, ε' is associated with energy storage through polarization, while ε'' and σ both contribute to electromagnetic loss in the medium.[7] As a result, measured transmission and reflection responses are affected not only by how strongly the medium stores electric energy, but also by how strongly it dissipates that energy.

In aqueous samples, water is typically the dominant constituent and plays a central role in determining the overall RF response. Water is a strongly polar

liquid whose dielectric behaviour varies with frequency. The complex permittivity of liquid water is known to be frequency-dependent, and over much of the RF range, is commonly described using Debye-type relaxation behaviour.[8]

When nanoparticles are suspended in a host medium, they perturb the effective dielectric properties of the composite suspension. From an effective-medium perspective, the bulk response of this heterogeneous material depends on the properties of the host medium, the properties of the inclusions, their geometry, and their concentration or volume fraction.[7]

Consequently, differences in the nanoparticle concentration will likely change the effective dielectric constant and dielectric loss of the aqueous solution. This may then be reflected in measured RF responses, although the extent of this effect can depend strongly on frequency, sample purification, and the surrounding ionic background.[9] The RF response is typically measured by its magnitude and phase offset, more details of which are discussed in section 4.2.

Prior work has experimentally investigated the radio and microwave response of nanoparticle colloids and aqueous gold nanoparticle dispersions, including dielectric-property retrieval from transmission measurements and RF absorption studies across kHz-to-GHz regimes. However, an exact prior match to this study’s measurement configuration as discussed in chapter 3 and chapter 4 was not identified in this review. [9, 10, 11]

2.3 UHF Radio Sensing

UHF generally refers to the radio-frequency range from 300 MHz to 3 GHz.[12] At much lower frequencies, electromagnetic waves may offer deeper penetration, but the associated wavelengths become longer and the antennas correspondingly larger, making compact exploratory measurement setups less convenient. At much higher microwave or millimeter-wave frequencies, interaction with materials may in some cases become more pronounced, but attenuation also tends to increase and the measurement setup can become more demanding in terms of hardware and calibration.[13] The UHF range is therefore an attractive target region as it balances against good penetration performance, and setup complexity, helped further by widely accessible commercial hardware such as software-defined radios and off-the-

CHAPTER 2. LITERATURE REVIEW

shelf antennas.

Another advantage of the UHF range is that it lies in the non-ionizing part of the electromagnetic spectrum.[14] As such, it is fundamentally different from ionizing imaging modalities such as X-rays. This does not mean that radio exposure should be treated carelessly, since RF systems are still subject to established exposure limits and safety guidelines, especially at higher transmitted powers.[15] However, for low-power experimental sensing, the non-ionizing nature of UHF supports its attractiveness as a candidate modality for non-invasive exploratory measurement.

Chapter 3

Simulating UHF Response

As an initial step to validating the hypothesis that differences in nanoparticle type and concentration level in the water-based solution should yield differences in the measured radio signal received when UHF radio waves are transmitted in proximity to the nanoparticle samples, a simulation study was first conducted using the Ansys HFSS software.[16] In this simulation, a frequency sweep of 1.5 - 2.5 GHz was performed to explore the samples' response across the UHF range.

3.1 Simulation Setup

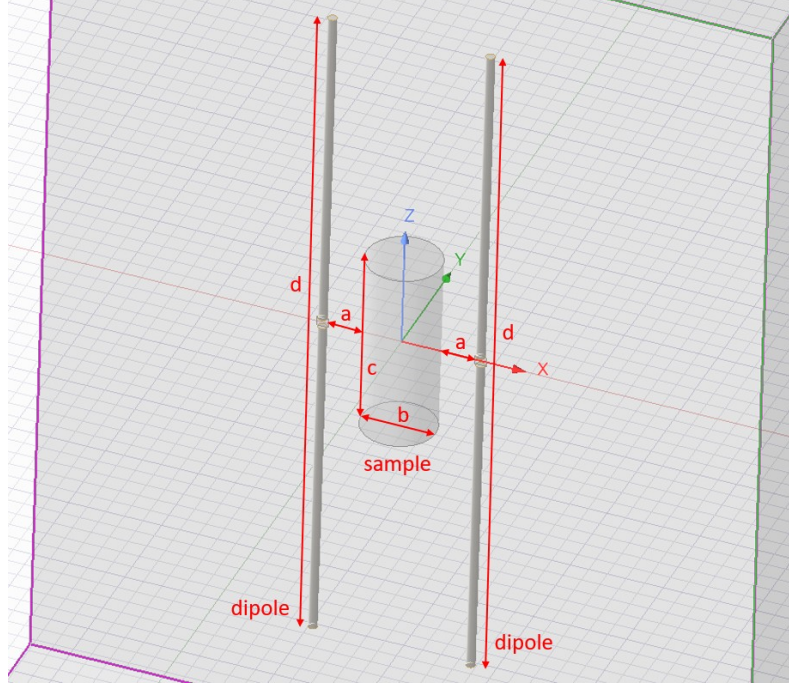
3.1.1 Transmission Coefficient S_{21}

The setup comprised 2 dipole antennas positioned on either side of the simulated nanoparticle sample (modelled as a cylinder). Both dipoles' material were modelled as perfect electric conductors (PEC). The sample's simulated materials are discussed in Section 3.1.3. The three components were defined within an airbox to model free-space radiation. Figure 3.1 illustrates the simulation setup.

The key dimensions of this simulation setup were as follows:

- Distance between each antenna and the sample (a): 4 mm
- Diameter of the sample cylinder (b): 8 mm
- Height of the sample cylinder (c): 20 mm
- Height of the dipole antenna (d): 73 mm

The relative proximity of the antennas to the sample in the simulation was due to the mesh count limit imposed on the free student edition version of the HFSS


 Figure 3.1: HFSS simulation setup for measuring S_{21} .

software. An object lying in the near field region of the antenna can directly perturb the localized electromagnetic field around the antenna rather than only interacting with the propagating radio wave.[17] As this simulation exercise was intended for initial exploration of whether the nanoparticle types and concentration levels may have effects on the radio signals, the sample being in the near field radiation region was deemed less of a concern in itself.

In a half-wave dipole design, the dipole antenna works optimally when its length is just under half a wavelength (roughly 150 mm at 2 GHz), giving stronger and more reliable readings. A slightly shorter physical length was used in the simulation runs because the field extended beyond the ends of the metal, so the dipole behaved electromagnetically as though it was a bit longer than it physically was.[18]

A frequency sweep from 1.5 GHz to 2.5 GHz was simulated in steps of 1 MHz, and the magnitude and phase of the transmission coefficient S_{21} were measured. The S_{21} refers to the complex phasor ratio of the received wave (at port 2) to the transmitted wave (from port 1), therefore the greater its magnitude, indicates stronger transmission through the sample. The technical details of the complex phasor formulation, and the derivation of the magnitude and phase for the given

phasor are provided in Section 4.2.

3.1.2 Reflection Coefficient S_{11}

Separately, to simulate radio wave reflection, a monostatic configuration was used, with a single dipole antenna positioned 7 mm from the sample. The dimensions of the sample and dipole antenna were similar to those used in the S_{21} simulation. Figure 3.2 illustrates the simulation setup.

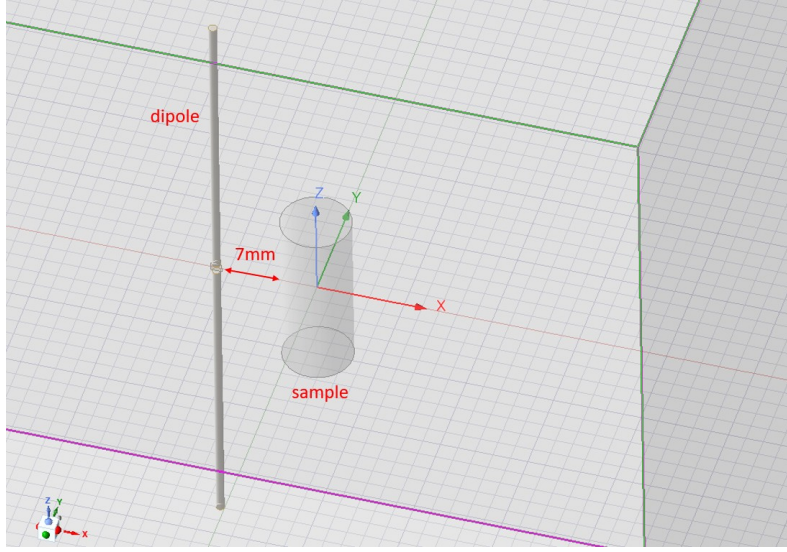


Figure 3.2: HFSS simulation setup for measuring S_{11} .

In this setup, the magnitude and phase of the reflection coefficient S_{11} were measured, where S_{11} is the complex phasor ratio of the reflected wave received to the transmitted wave at the same port 1. A higher magnitude measured generally indicates a stronger reflection of the radio wave from the sample.

3.1.3 Sample Material

There is no library material in the HFSS software that models the water-based nanoparticle samples intended for physical testing later. The sample's material model therefore had to be specified by its relative permittivity and dielectric loss tangent values for every radio frequency value in 1 MHz steps.

The dielectric loss tangent represents how much of the electromagnetic energy in a dielectric is dissipated as loss rather than stored when an alternating electric

CHAPTER 3. SIMULATING UHF RESPONSE

field is applied. In complex-permittivity form, it is defined as:

$$\tan \delta = \frac{\varepsilon''}{\varepsilon'} \quad (3.1)$$

where ε'' is the imaginary part, and ε' is the real part of the permittivity.[11]

For a mixed medium like the nanoparticle solution, the Maxwell-Garnett model can be utilized to model its permittivity as long as the wavelength of the radio wave is much longer compared with the size of the particles and their separation distances.[9] These are fulfilled in this simulation exercise with the target wavelength being in the region of 150 mm at 2 GHz, the nanoparticle size at 10 nm and a minimum concentration of 0.1 mg/mL.

$$\varepsilon_{\text{eff}} = \varepsilon_1 \frac{\varepsilon_m + 2\varepsilon_1 + 2q(\varepsilon_m - \varepsilon_1)}{\varepsilon_m + 2\varepsilon_1 - q(\varepsilon_m - \varepsilon_1)} \quad (3.2)$$

where ε_m and ε_1 are the complex permittivities of the nanoparticles and the surrounding medium respectively, q is the volume fraction of the metal, and ε_{eff} is the effective dielectric constant of the composite medium.

The permittivity of water was computed at each frequency point in the sweep using the temperature-dependent double-Debye model reported by Liebe, Hufford, and Manabe for pure water below 1 THz.[8] For each frequency point, the modified temperature variable was first evaluated for $T = 22^\circ\text{C}$:

$$\theta = 1 - \frac{300}{273.15 + T} \quad (3.3)$$

The following temperature-dependent parameters were then evaluated:

$$\varepsilon_0 = 77.66 - 103.3\theta, \quad (3.4)$$

$$\varepsilon_1 = 0.0671\varepsilon_0, \quad (3.5)$$

$$\varepsilon_2 = 3.52 + 7.52\theta, \quad (3.6)$$

$$\gamma_1 = 20.20 + 146.4\theta + 316\theta^2, \quad (3.7)$$

$$\gamma_2 = 39.8\gamma_1. \quad (3.8)$$

These parameters were then substituted into the double-Debye expression for the complex permittivity of water, so that a frequency-specific value of water permittivity was obtained for every point in the sweep:

$$\varepsilon_{m,\text{water}}(f) = \frac{\varepsilon_0 - \varepsilon_1}{1 - i\left(\frac{f}{\gamma_1}\right)} + \frac{\varepsilon_1 - \varepsilon_2}{1 - i\left(\frac{f}{\gamma_2}\right)} + \varepsilon_2 \quad (3.9)$$

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where f , γ_1 , and γ_2 are expressed in GHz.

The complex permittivity of gold as a function of frequency was computed using the Drude free-electron model, with the Drude plasma energy and relaxation time taken from Olmon *et al.*[19] Specifically, the computation used a reported plasma energy of $E_p = 8.45$ eV and a relaxation time of $\tau_D = 14$ fs. The corresponding plasma angular frequency ω_p and relaxation rate Γ were then evaluated as follows:

$$\omega_p = \frac{E_p}{\hbar} \quad (3.10)$$

where $E_p = 8.45 \times 1.602176634 \times 10^{-19}$ J, and $\hbar = 1.0545718 \times 10^{-34}$ J · s. [20]

The corresponding relaxation rate was obtained from the relaxation time as:

$$\Gamma = \frac{1}{\tau_D} \quad (3.11)$$

For each sweep frequency f , the angular frequency was obtained as:

$$\omega = 2\pi f \quad (3.12)$$

The complex permittivity of gold was then evaluated at each frequency point using:

$$\varepsilon_{m,\text{gold}}(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\Gamma\omega}. \quad (3.13)$$

By finding the complex permittivities for both gold and water respectively at each frequency point, the Maxwell-Garnett effective-medium calculation could then be used to find the effective complex permittivity of the gold nanoparticle solution for each given concentration level and frequency point. This could then be used to derive the dielectric loss tangent to complete the sample material modelling for the HFSS software.

In this simulation exercise, 5 gold nanoparticle solution materials were modelled after the following concentration levels per the actual samples that will subsequently be used in later physical experiments: 0.01, 0.02, 0.04, 0.05 and 0.08 mg/mL. These sample models may interchangeably be referred to as au01, au02, au04, au05 and au08 respectively in the subsequent discussions of this section.

3.2 Simulation Results

3.2.1 Transmission Coefficient S_{21}

Referring to Figure 3.3, several observations could be made about the magnitude of the S_{21} ($|S_{21}|$), measured in dB, across the frequency range of 1.5 - 2.5 GHz, and for the different concentration levels:

- $|S_{21}|$ for each concentration level showed very subtle differences, roughly in the order of 10^{-6} db and cannot be easily resolved visually by the graph.
- $|S_{21}|$ appeared to peak in the region of 1.8 - 1.9 GHz across the various concentration levels.

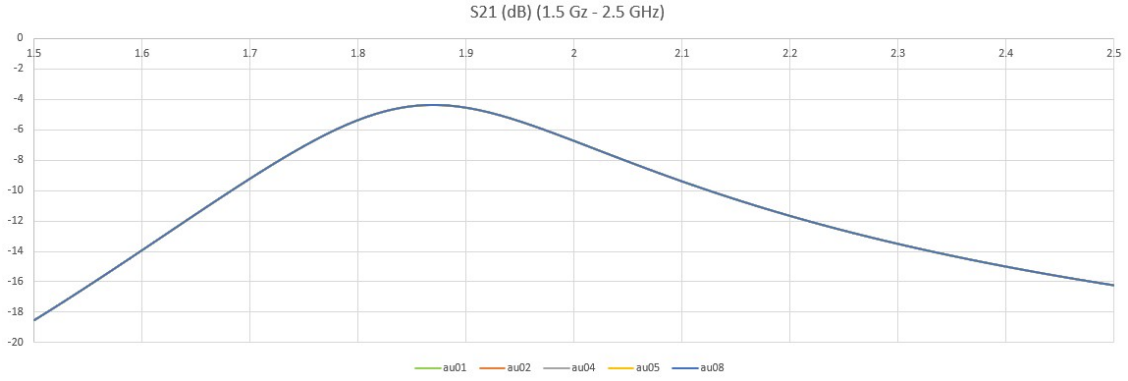


Figure 3.3: $|S_{21}|$ (y-axis) of gold nanoparticle samples for 0.01, 0.02, 0.04, 0.05 and 0.08 mg/mL concentration level respectively, across 1.5 - 2.5 GHz (x-axis).

It was found that the relationship between the concentration level and $|S_{21}|$ was not consistently positive or negative across the frequency sweep range. But it was also found that regardless of the relationship direction, it was always monotonic: Meaning that if the relationship was negative for example, higher concentration levels always led to lower measured $|S_{21}|$ values.

We can take advantage of this fact to find the relationship between $|S_{21}|$ and the concentration level by taking the difference between the measured $|S_{21}|$ of a higher concentration level against that of a lower concentration level across the frequency sweep range: A positive difference meant a positive relationship and vice versa for a negative difference.

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Thus the relationship between $|S_{21}|$ and the concentration level can be described as positive between 1.5 GHz to roughly 1.82 GHz, and negative at higher frequencies, from Figure 3.4, which shows the difference between $|S_{21}(au08)|$ against $|S_{21}(au01)|$ across 1.5 - 2.5 GHz.

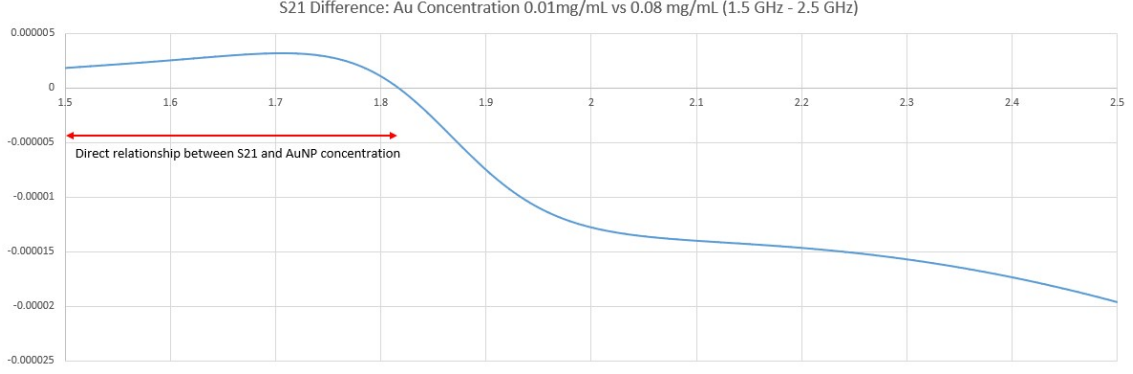


Figure 3.4: $\Delta|S_{21}| = |S_{21}(au08)| - |S_{21}(au01)|$ (y-axis), for 1.5 - 2.5 GHz (x-axis).

Referring to Figure 3.5, similar to the case of $|S_{21}|$, the phase of the S_{21} ($\angle S_{21}$), measured in degrees, showed only very subtle difference across the various concentration levels, in the rough order of 10^{-5}° . In general, $\angle S_{21}$ appeared to have a negative relationship with frequency.

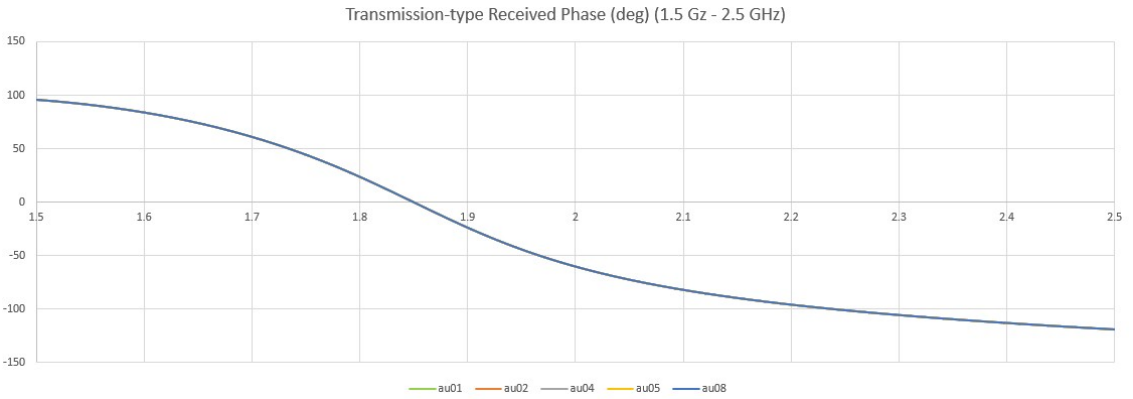


Figure 3.5: $\angle S_{21}$ (y-axis) of gold nanoparticle samples for 0.01, 0.02, 0.04, 0.05 and 0.08 mg/mL concentration level respectively, for 1.5 - 2.5 GHz (x-axis).

The relationship between $\angle S_{21}$ and concentration level was also found to be monotonic. Using similar subtraction trick as with the case of $|S_{21}|$, Figure 3.6 revealed that $\angle S_{21}$ and the nanoparticle solution's concentration level shared a negative relationship across the 1.5 - 2.5 GHz sweep range.

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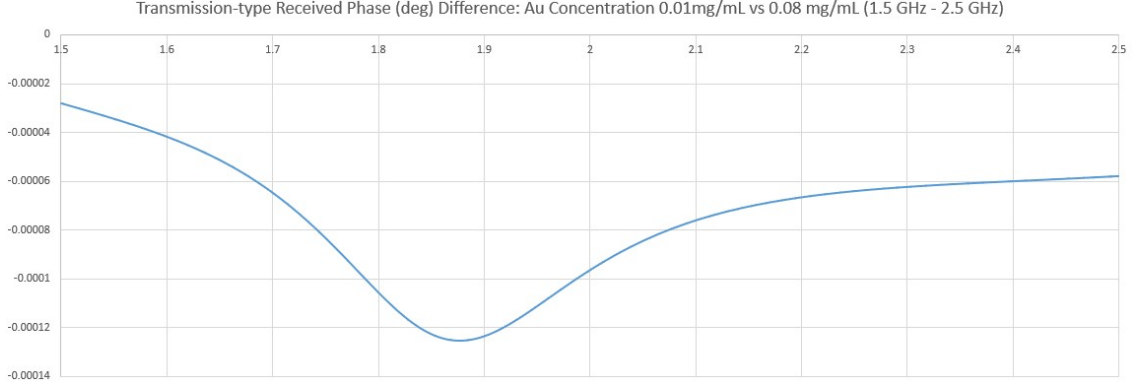


Figure 3.6: $\Delta\angle S_{21} = \angle S_{21}(au08) - \angle S_{21}(au01)$ (y-axis), for 1.5 - 2.5 GHz (x-axis).

3.2.2 Reflection Coefficient S_{11}

Figure 3.7 showed an RF reflection trend consistent with that observed for the transmission coefficient $|S_{21}|$. Since $|S_{21}|$ reached its maximum in the 1.8 - 1.9 GHz region, indicating that the radio wave was transmitted through the nanoparticle solution more readily in that band, $|S_{11}|$ was correspondingly lowest over the same frequency range.

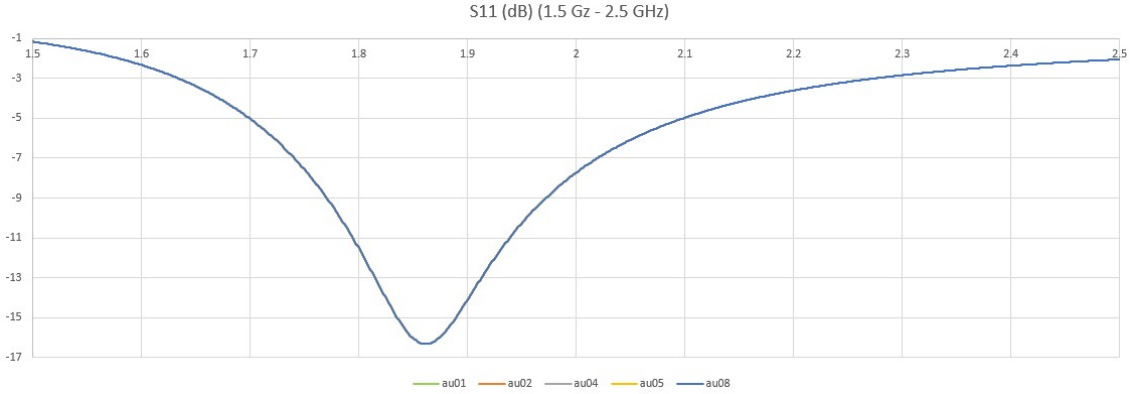


Figure 3.7: $|S_{11}|$ (y-axis) of gold nanoparticle samples for 0.01, 0.02, 0.04, 0.05 and 0.08 mg/mL concentration level respectively, for 1.5 - 2.5 GHz (x-axis).

Because S_{11} was also found to be monotonic with concentration levels, allowing us to apply the subtraction trick as well, Figure 3.8 showed that S_{11} had a negative relationship with the concentration levels between 1.5 and roughly 1.86 GHz, and then a positive one for higher frequencies.

Interestingly, Figure 3.8 also showed that the greatest extent of the negative

CHAPTER 3. SIMULATING UHF RESPONSE

relationship occurred roughly in the region of 1.82 GHz, and the greatest extent of the positive relationship was observed at roughly 1.9 GHz.

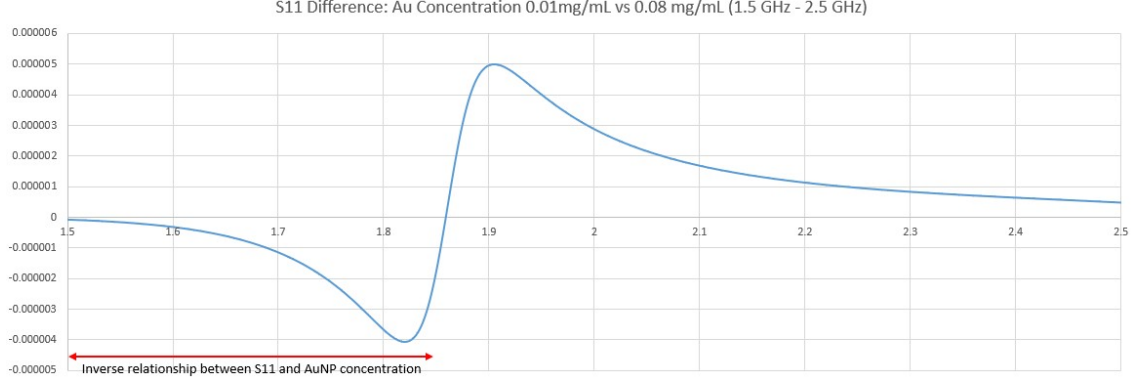


Figure 3.8: $\Delta|S_{11}| = |S_{11}(au08)| - |S_{11}(au01)|$ (y-axis), for 1.5 - 2.5 GHz (x-axis).

Figure 3.9 showed that $\angle S_{11}$ was initially negative, decreasing with frequency and reaching a local minimum at roughly 1.75 GHz, before rising steeply becoming positive from roughly 1.88 GHz onwards, reaching a local maximum at close to 2 GHz, after which it tapered off gradually.

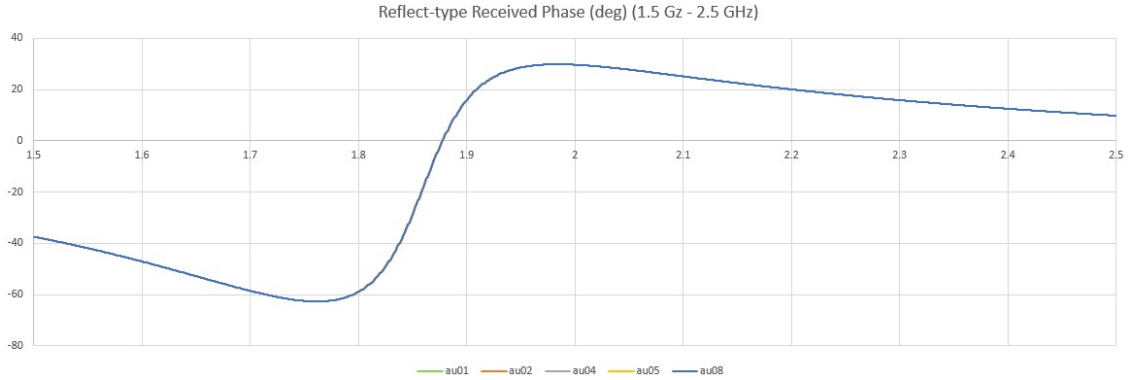


Figure 3.9: $\angle S_{11}$ (y-axis) of gold nanoparticle samples for 0.01, 0.02, 0.04, 0.05 and 0.08 mg/mL concentration level respectively, for 1.5 - 2.5 GHz (x-axis).

$\angle S_{11}$ proved to be monotonic in relationship with concentration levels, so we can again employ the same subtraction trick to show from Figure 3.10 that aside from the frequency range of roughly 1.76 - 1.96 GHz which exhibited a positive relationship, the relationship between $\angle S_{11}$ and the concentration level can be described as negative, and smaller in extent than the peak of the positive relationship phase within the 1.5 - 2.5 GHz frequency sweep range.

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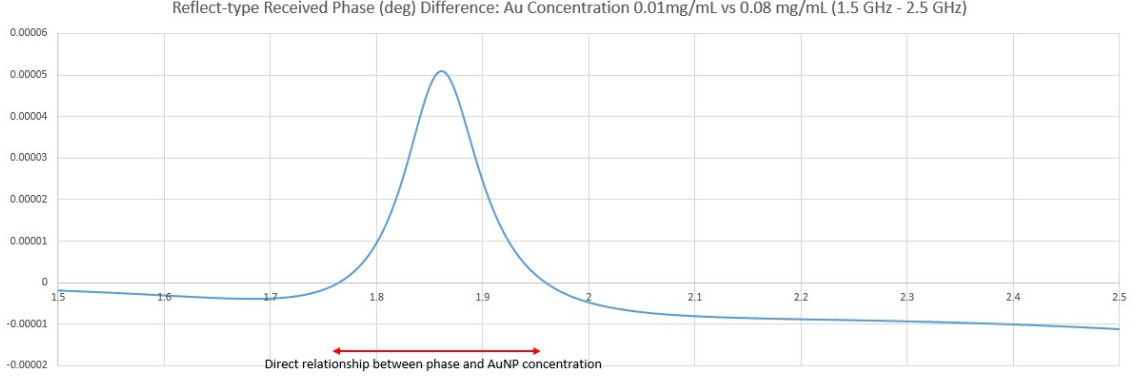


Figure 3.10: $\Delta\angle S_{11} = \angle S_{11}(au08) - \angle S_{11}(au01)$ (y-axis), for 1.5 - 2.5 GHz (x-axis).

3.2.3 Summary

The simulation data suggested that 1.8 GHz would be a suitable target frequency for the subsequent physical experiments seeking to validate measurable differences in magnitude and phase of the UHF wave as it interacts with nanoparticle solutions with varying concentration levels for the following reasons:

1. The signal-to-noise ratio would be close to maximum for measuring the transmission coefficient $|S_{21}|$, improving the physical measurement consistency.
2. It avoids the transition from positive and negative relationship between $|S_{21}|$ and concentration levels.
3. It is close to the frequency range where the negative relationship between $\angle S_{21}$ and concentration levels is at its greatest extent.
4. Although a higher value of $|S_{21}|$ would likely trade-off against the reflection coefficient $|S_{11}|$ and affect its signal-to-noise ratio, it appears that the extent of the negative relationship between $|S_{11}|$ and concentration levels is larger and would hopefully offset a weaker signal-to-noise ratio.
5. It is close to the frequency range where the positive relationship between $\angle S_{11}$ and concentration levels is at its greatest extent.

Based on the simulated UHF response at 1.8 GHz, the expected relationship between concentration level and the magnitude, and the phase offset of the transmission and reflection coefficient is summarized in Table 3.1.

Table 3.1: Simulated relationship between the measured parameter and concentration level for the 1.8 GHz UHF setup.

Coefficient Type	Parameter	Trend with Concentration
Transmission	Magnitude	Positive
	Phase	Negative
Reflection	Magnitude	Negative
	Phase	Positive

At UHF range, both gold and silver are both very good conductors and are therefore expected to exhibit broadly similar RF response patterns.[21] Although silver should exhibit higher conductivity than gold because of its higher conductivity, simulation using gold should still be a reasonable initial baseline for understanding the expected response of silver.

3.3 Initial Physical Validation

An initial validation was made by setting up a physical experiment measuring the transmission of the radio wave through the sample. The technical details of the Software-Defined Radio (SDR), radio signal processing and how the measurements were derived are discussed in chapter 4.

By placing the sample between the two antennas, the radio waves can be transmitted from one antenna and then received by the other antenna and measured after passing through the sample. Both antennas were mounted 200 mm apart, with the sample placed in-between (i.e. 100 mm away from either antennas), so that they were not within the near-field zone, which avoids extreme geometry sensitivity such as from cable-flexing and even slight geometry differences from the sample’s placement.

The edge of the near-field zone is typically calculated as $\frac{nD^2}{\lambda}$, where n is usually 2, D is the diameter of the smallest sphere that encloses the antenna’s radiating parts, and λ is the wavelength.[22] The Siretta Delta 1D antenna, suitable for 1-2 GHz frequency range, was used in the experiment.[23] With $n = 2$, $D = 51$ mm and $\lambda = 166.55$ mm (based on 1.8 GHz), the near-field distance was estimated to be 31.2 mm, which is less than the physical spacing used in the experimental set-up. The coaxial cables were kept relatively short at 300 mm long to minimise cable-

related signal loss.[24] For each water and nanoparticle sample, a 300 μL volume was pipetted into a viewplate, which was then placed between the antennas during measurement.

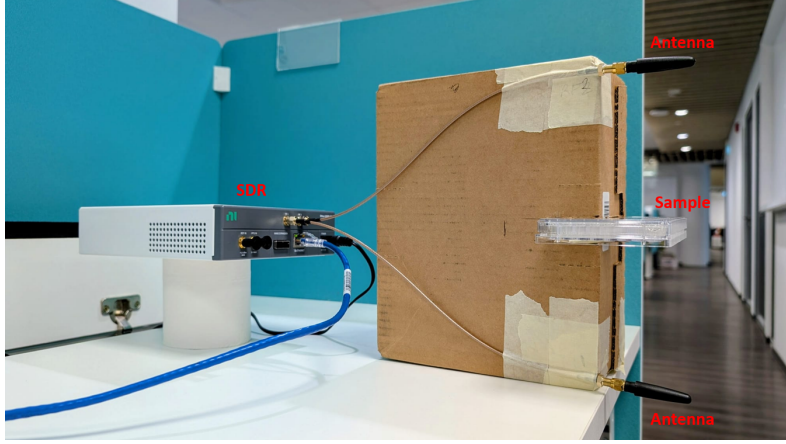


Figure 3.11: The physical measurement setup.

A frequency sweep between 1.5 and 2.5 GHz was performed in steps of 1 MHz. For each frequency point, the mean of the magnitude of the radio wave (measured in dB) transmitting through the sample was calculated for 0.8 seconds of sampling at a rate of 1 MHz (800,000 readings per frequency point). Between each frequency point's measurement set, a 0.2 second measurement gap was maintained to avoid data stability issues observed during the transition between frequency points.

Figure 3.12 shows the magnitude of the radio wave transmitting through various samples (air, water and the different gold nanoparticle samples with differing concentration levels). The data suggested that:

- The trend of the magnitude broadly followed that of the HFSS simulations for $|S_{21}|$, peaking in the region of 1.8 GHz, validating the subsequent experimental focus on that particular frequency.
- Also, as suggested by the simulations, the measured magnitude differences were small, indicating the need for improvements to the experimental approach, that could suppress noise and improve sensitivity to the sample-induced UHF response.

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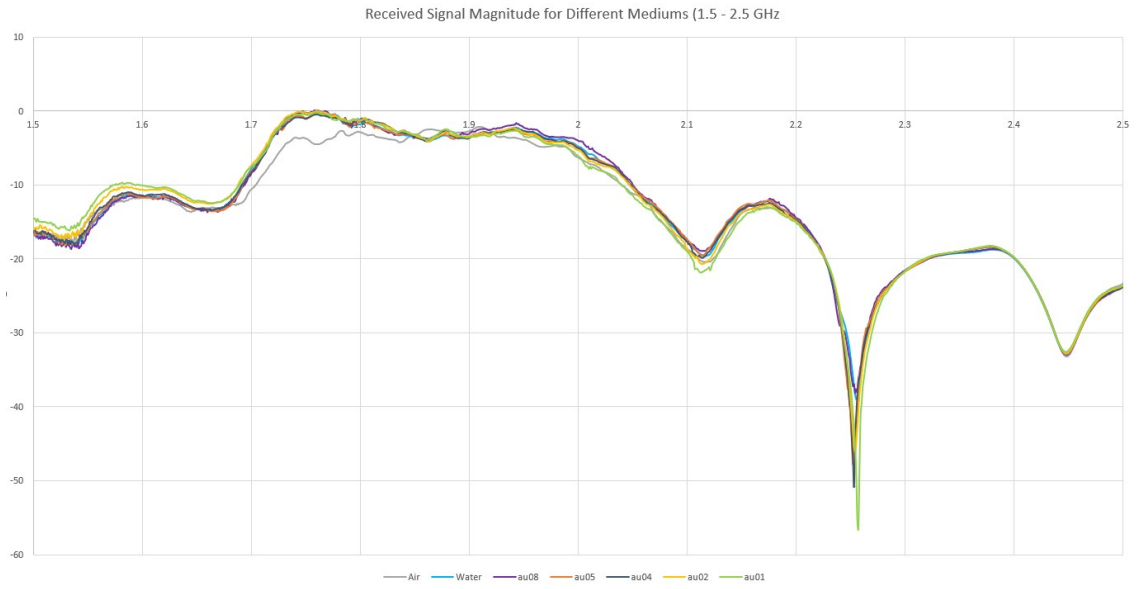


Figure 3.12: Magnitude of the radio wave (y-axis) transmitted through the various samples, for 1.5 - 2.5 GHz (x-axis).

Chapter 4

Measurement Methodology

4.1 Experimental Setup

An SDR, an Ettus N210 with an SBX-40 daughterboard in this case, was used to transmit and receive the radio waves.[25] The transmitting and receiving RF antennas were connected to the SDR through their respective coaxial cables to avoid potential interference with the SDR itself.

The initial physical measurement validation discussed in section 3.3 had indicated the need for improvements to the experimental approach to enhance the measurement sensitivity to the sample-induced RF response. Changes were therefore made to the setup:

1. The omnidirectional Siretta Delta 1D antennas were replaced by 100×100 mm directional patch antennas that focus the radio wave within a beamwidth ranging between 24° and 48° .
2. To avoid the near-field zone, it was calculated that the sample should not be placed nearer than 250 mm.
3. Because the gap between the antennas and the sample had grown, the existing 300 mm cables were extended to 600 mm.
4. To maximize the sample volume for better RF response, the liquid samples were no longer pipetted into the viewplate and instead left in their respective microcentrifuge tubes, with the default sample volume of 1 mL.

4.1.1 Transmission Measurement Setup

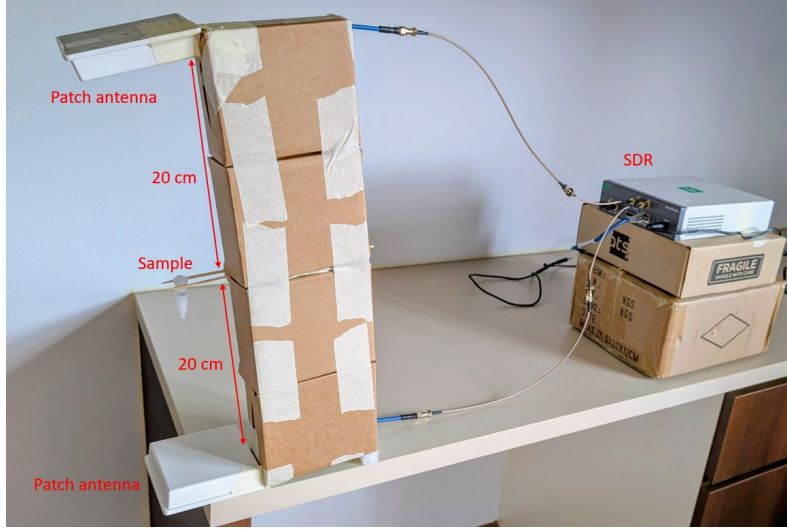


Figure 4.1: The transmission measurement's physical setup.

Due to material constraint, the sample could be placed only 200 mm from the antennas instead of the intended 250 mm for the transmission measurement setup.

4.1.2 Reflection Measurement Setup

A bistatic configuration was used for the reflection measurement setup, where the transmitting and receiving antenna were mounted adjacent to each other with a physical separation, and radiating the radio waves outwards in the same general direction.

The relevant physical dimensions of the setup are illustrated in Figure 4.3 for reference. In particular, the antennas were setup so that at the narrowest beamwidth of 24° , the sample will still be within the antennas' beamwidth at the distance of 250 mm away from the main antennas assembly.

4.2 Signal Processing

In general, a radio wave can be mathematically represented by the formula:

$$x(t) = A \cos(2\pi ft - \phi) \quad (4.1)$$

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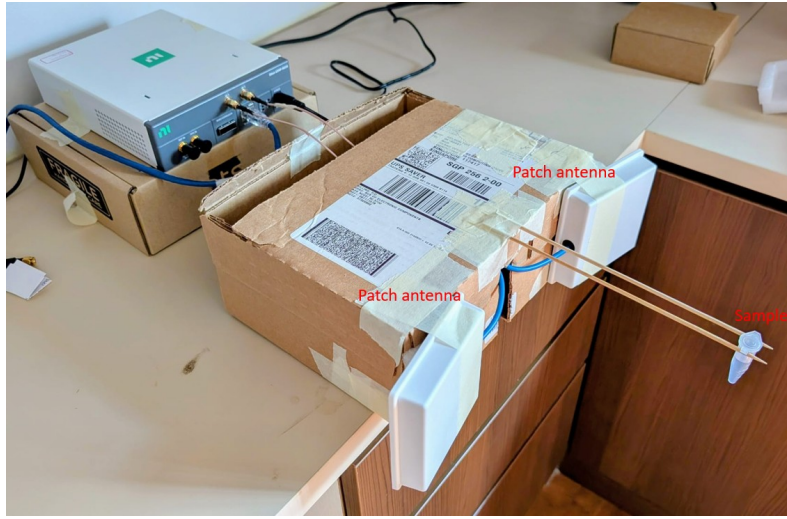


Figure 4.2: The reflection measurement's physical setup.

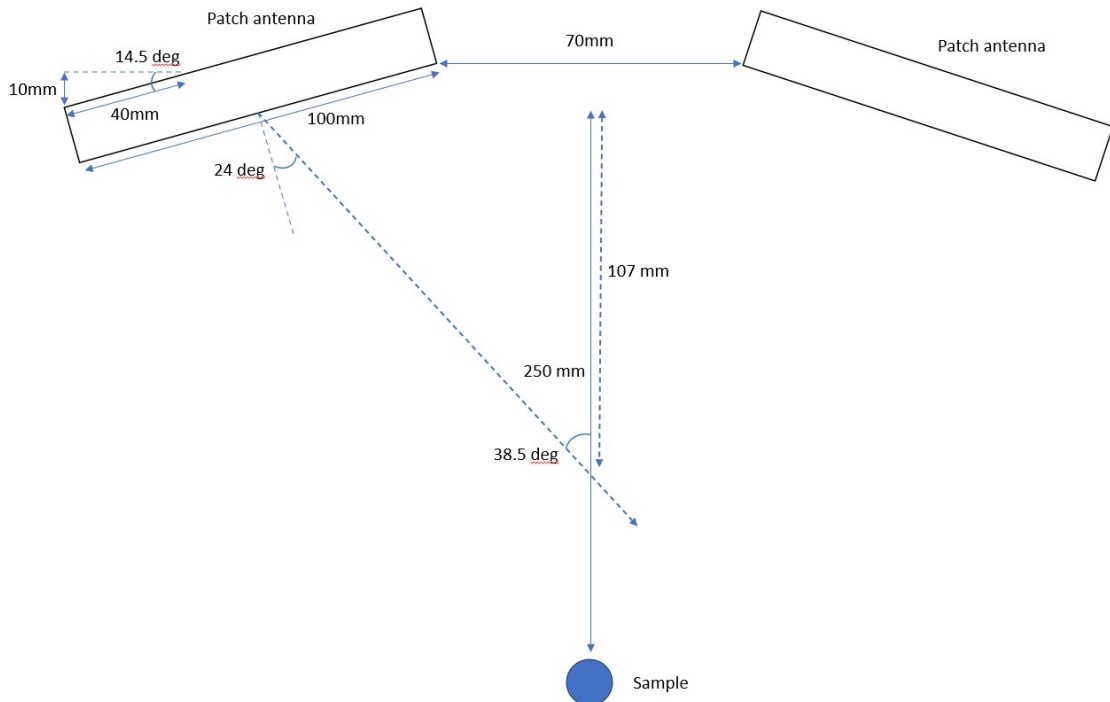


Figure 4.3: The reflection measurement's physical dimensions (top-down view, not exactly drawn to scale).

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where $x(t)$ is the signal at time t , A is the amplitude, f is the frequency, t is the time elapsed, and ϕ is the phase offset of the wave (i.e. how much the wave is shifted left or right in time compared to a reference wave).

Based on the trigonometric identity: $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$, we can represent the same wave as the sum of two separate constituent waves that are in quadrature, or 90 degrees out of phase:

$$x(t) = I \cos(2\pi ft) + Q \sin(2\pi ft) \quad (4.2)$$

where $I = A \cos \phi$ and $Q = A \sin \phi$.

Also based on the trigonometric identity $\sin^2 \phi + \cos^2 \phi = 1$, for a given phase shift ϕ , the amplitude of the wave, A , can be found by calculating the magnitude of the phasor vector formed by the I and Q components, with I as the x-axis and Q as the y-axis on the phasor diagram:

$$A = \sqrt{I^2 + Q^2} \quad (4.3)$$

Further, since $\tan \phi = \frac{\sin \phi}{\cos \phi}$, then:

$$\phi = \text{atan2}(Q, I) \quad (4.4)$$

Therefore, a signal of the radio wave with frequency f , can have its amplitude A and its phase shift ϕ derived by the values I and Q instead, which is indeed output by the SDR as measurement readings of the radio wave signals.

It happens that a complex number is expressed as: $c = a + jb$, where c is a complex number comprising a real component a , and an imaginary component b , with $j = \sqrt{-1}$. By plotting the imaginary component against the real component, a complex number can be also be represented by a vector with magnitude $\sqrt{a^2 + b^2}$ and angle $\tan^{-1} \frac{b}{a}$.

By treating I as a real component, and Q as an imaginary component, we can express an I-Q vector in polar form $z = A(\cos(\phi) + j \sin(\phi))$, which is commonly written as $z = I + jQ$ and referred to as a phasor. Then using Euler's formula, we can express in exponential form: $z = Ae^{j\phi}$. [26]

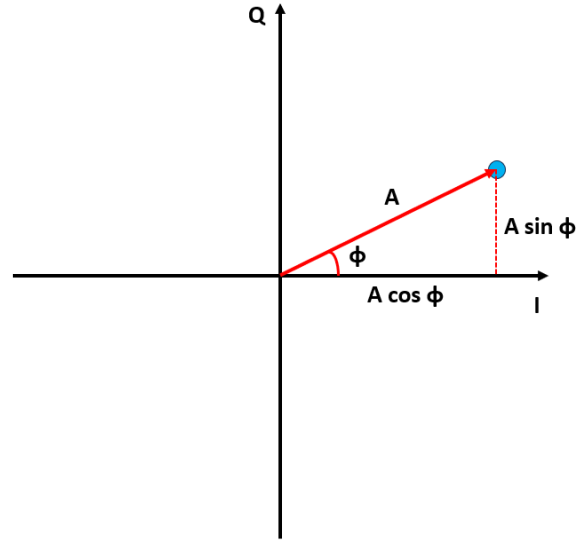


Figure 4.4: A phasor diagram represents a signal by plotting Q vs I , producing a vector with magnitude A and angle ϕ .

4.3 RF Sampling

When taking sample measurements of a radio wave's signals, the minimum sampling rate must be at least twice the radio wave frequency to avoid aliasing. This is referred to as the Nyquist Rate.[27] Direct sampling of UHF signals is therefore challenging due to hardware and computational hardware bottlenecks such as the SDR's ADC (Analog-to-Digital Converter) and the bandwidth of the ethernet connection to the host computer. In particular, the Ettus N210's specifications indicate a sampling rate limit of 50 MHz, making direct sampling of UHF signals infeasible. [25]

To solve this, the SDR does not directly generate the target sinusoid in software. Instead, it generates a low-frequency waveform centred around 0 Hz by default, referred to as the baseband. The SDR then uses mixers to translate this baseband waveform up to the target frequency for transmission, and then translate the received RF signal back down to the baseband for sampling. This process is referred to as direct conversion[28], allowing us to take readings of the received signal at a sampling rate of 1 MHz.

In practice, the process of direct conversion results in LO (Local Oscillator) "leakage" which is caused by frequency mixing resulting in additional energy created.

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This frequently manifests as a spike in relative gain at the baseband frequency of 0 Hz, causing inaccuracies to sample readings.

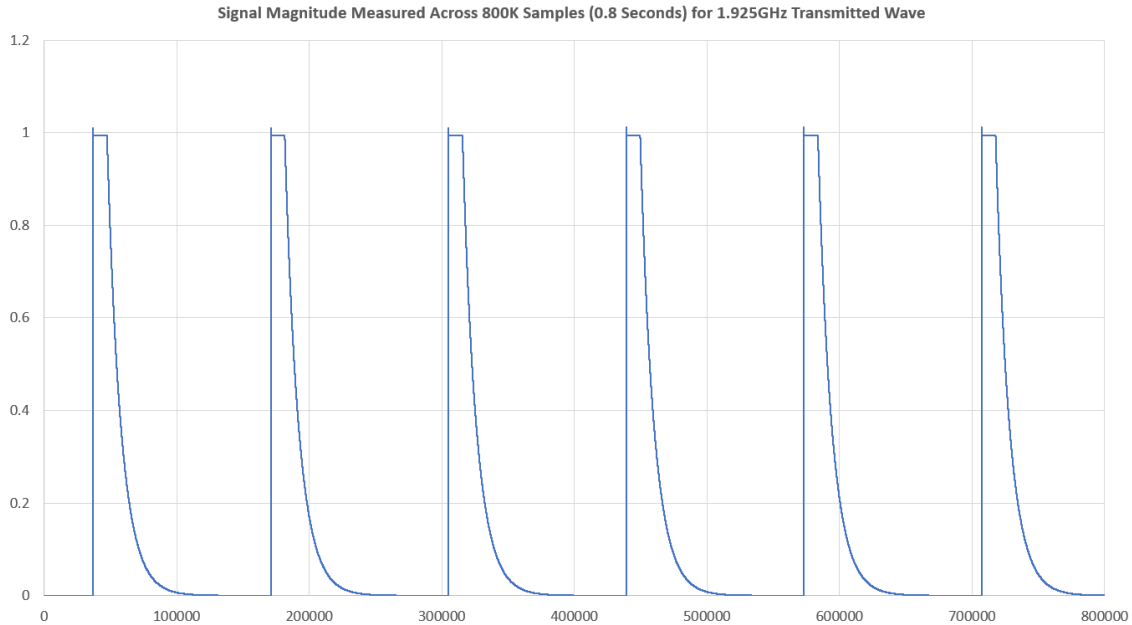


Figure 4.5: Spikes in signal magnitude caused by LO leakage at baseband frequency 0 Hz were observed when taking continuous measurement (for 0.8 seconds corresponding to 800,000 samples at a sampling rate of 1 MHz).

To mitigate the issue of LO leakage, a frequency offset was applied to the baseband to avoid 0 Hz. Experiments confirmed that choosing an arbitrary baseband offset frequency of 97.65625 Hz provided a very good balance between eliminating the LO leakage spikes by staying far enough from 0 Hz, and allowing us to still respect the Nyquist Rate with the intended sampling rate of 1 MHz.

4.4 Standard Sample Bracket Measurement

RF signals are often affected by external RF noise and interference from the surrounding environment, of both natural and man-made contributions. These include background cosmic radiation, unwanted emissions from other RF systems operating in the same or nearby bands.[29], and even nearby human activities.[30]

We can try to minimise the interference, especially on the latter two sources, by sanitising the immediate physical environment through avoiding active RF sources (e.g. switching phone and laptop to flight mode during experiments), and imple-

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menting the experimental setup further from high human footfall areas. But such an approach itself neither fully eliminates all interference, nor is it scalable beyond the experimental set-up into eventual real-world applications.

A common technique used to mitigate against background noise is standard sample bracketing, where the measurement steps can be described broadly as follows:[31]

1. Take a reference measurement (without sample) x_{r1} , at time t_{r1} .
2. Take the measurement with the sample x_s , at time t_s .
3. Take a second reference measurement (without sample) x_{r2} , at time t_{r2} .
4. Calculate a linear time-interpolated reference measurement between both reference measures: $x_{ref} = x_{r1} + \left(\frac{t_s - t_{r1}}{t_{r2} - t_{r1}} \times (x_{r2} - x_{r1})\right)$.
5. Normalise the sample measurement against the reference measurement by finding the ratio: $r_{s,ref} = \frac{x_s}{x_{ref}}$.

Although the standard sample bracketing technique was developed to control for background noise, it works best when the background noise is stable throughout the measurement phase. To achieve this, the experimental setup was physically sited at an environment with minimal human activities, and the measurement duration is minimised by taking the bracketing measurements immediately time-adjacent to the sample measurement, reducing the potential exposure to background noise spikes skewing the measurements.

Measurements using the SDR were done in the following steps:

1. Begin a continual RF transmission without any sample first (for first bracket reading).
2. Ignore the received signals for the first 5 seconds to allow the readings to stabilise.
3. Record the phasors received for 1 second at 1 MHz sampling rate. This set of 1,000,000 readings forms the first bracket reading.

4. Whilst continually transmitting the radio wave, place the sample between the antennas, ignoring the received signals during this "sample-swap" phase.
5. With the sample secured, repeat Steps 2 and 3 to record the received signals with the sample.
6. Remove the sample, again ignoring the received signals through this "sample-swap" phase.
7. Repeat Steps 2 and 3 to record the received signals without the sample (second bracket reading).

4.5 Data Processing

The aim is now to output the standardised sample's magnitude and phase offset of the RF signal passing through the sample, for the given transmitting frequency.

4.5.1 Removing Baseband Offset

The SDR returns each received signal reading as a phasor. Each phasor describes a transmitted wave that had been received and modulated to the baseband offset frequency of 97.65625 MHz to support sampling, and to avoid LO leakage. Modulation of a wave to a desired baseband offset frequency f_{off} is achieved through multiplying the wave signal by the complex exponential $e^{j2\pi f_{off}t}$, so that in exponential form: [26]

$$x_{off}(t) = x(t)e^{j2\pi f_{off}t} \quad (4.5)$$

To demodulate the wave therefore, each phasor z is multiplied by the conjugate of the complex exponential: $e^{-j2\pi f_{off}t}$, so that the transmitted's wave's phasor can be recovered.

4.5.2 Removing Carrier Frequency Offset

In theory, eliminating the baseband offset should bring the signal back to baseband frequency 0 Hz and the phase shift of the sample derived from the phasors should not have a trend over time. In practice however, a slight trend to the phase

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offset over time was observed, indicating a slow background phasor rotation even after offset frequency demodulation.

The impact to the phasors is a time-variant "drifting" of the phase offset even at theoretical baseband 0 Hz. This drift is referred to as the Carrier Frequency Offset (CFO), which occurs primarily due to hardware limitations such as the fact that the receiver and the transmitter have their respective LOs which can be very slightly out of sync with each other.[32]

To estimate, and then mitigate the CFO, we first sum the phasors across 0.01 second windows, for each bracket segment (i.e. pre-sample bracket, sample, post-sample bracket). The 1 second segment length used in this set-up therefore yields 100 summed phasors. Summing the phasors improves the signal-to-noise ratio for generally coherent signals, and therefore reduces the relative effect of noise on the measured phase of the summed phasors.[33]

A least-squares best-fit line for the phase is then estimated for each bracket segment b , yielding the slope representing the estimated phase drift per second. This allows us to estimate the CFO, f_b , which is then applied as an additional demodulation step by similarly multiplying the phasors by their respective bracket's estimated complex exponential conjugate: $e^{-j2\pi f_b t}$.

Therefore to summarise, two demodulation steps have now been applied to the phasors for their respective bracket segment b :

$$z_{1,b}(t) = z_b(t)e^{-j2\pi f_{\text{off}}t} \quad (4.6)$$

$$z_{2,b}(t) = z_{1,b}(t)e^{-j2\pi f_b t} \quad (4.7)$$

4.5.3 Standard Sample Bracketing

Across each 1 second bracket segment, the twice-demodulated complex phasors within are summed to form the respective aggregated phasors for each segment: Z_{a1} for the pre-sample bracket, Z_s for the sample itself, and Z_{a2} for the post-sample bracket.

We then derive the time-interpolated reference for normalization:

$$Z_{ref} = Z_{a1} + \left(\frac{t_s - t_{a1}}{t_{a2} - t_{a1}} \times (Z_{a2} - Z_{a1}) \right) \quad (4.8)$$

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where t_{a1} and t_{a2} are the time t mid-way within the respective bracketing segments before and after the sample segment, and t_s is the time t mid-way within the sample segment.

We can then find the standardized sample phasor: $\bar{Z} = Z_s/Z_{ref}$, and derive the standardized magnitude and phase offset of the sample phasor as measured and aggregated across 1 second. As the magnitude is usually reported in dB units, the magnitude of $\bar{Z}$ is expressed as:

$$\bar{Z}_{dB} = 20 \log_{10} |\bar{Z}| \quad (4.9)$$

Chapter 5

Experimental Results

For each concentration level, the standard sample bracketing measurement of the magnitude and phase was repeated 100 times per the methodology discussed in chapter 4. Subsequent sections to this chapter discuss the data observed for the transmission and reflection measurements, for gold and silver nanoparticles each.

5.1 Transmission Measurements

The summary statistics for the transmission measurements of gold and silver nanoparticles are shown in Table 5.1 and Table 5.2 respectively. These values are reported as mean $\pm$ standard deviation.

Table 5.1: Transmission measurement summary statistics for gold at each concentration level.

Concentration (mg/mL)	Magnitude (dB)	Phase (deg)
0.00	0.186 ± 0.069	-2.981 ± 0.390
0.01	-0.140 ± 0.118	0.053 ± 0.732
0.02	0.242 ± 0.045	4.477 ± 0.397
0.04	0.066 ± 0.057	-0.761 ± 0.380
0.05	0.179 ± 0.122	-1.092 ± 0.848
0.08	0.093 ± 0.032	-2.329 ± 0.295

Figure 5.1 and Figure 5.2 show the scatterplot of the standard sample bracketed magnitude vs phase for the transmission measurements of the gold and silver nanoparticles respectively.

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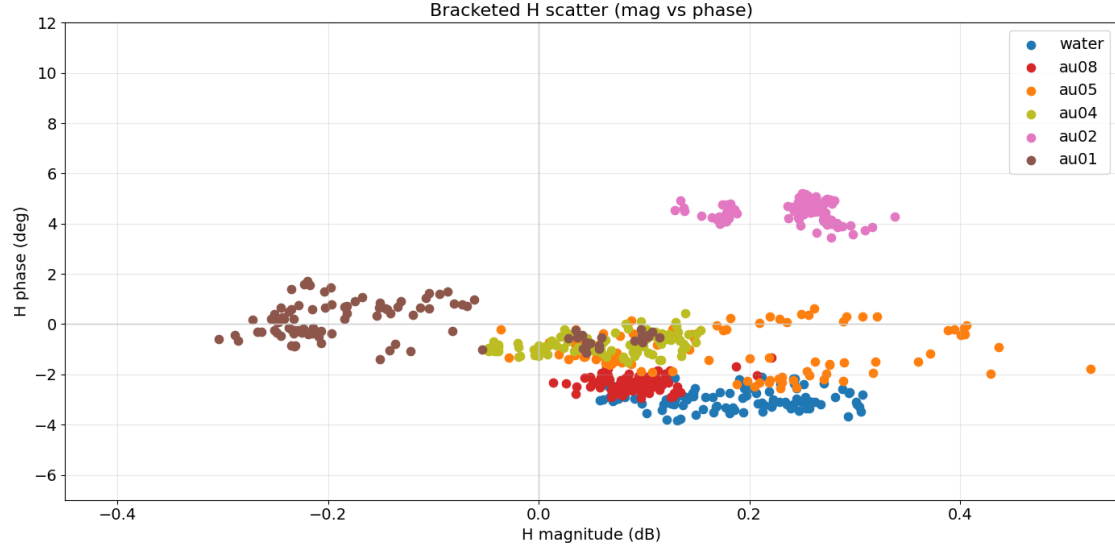


Figure 5.1: Scatterplot of bracketed magnitude versus phase for gold nanoparticles, for the transmission measurements.

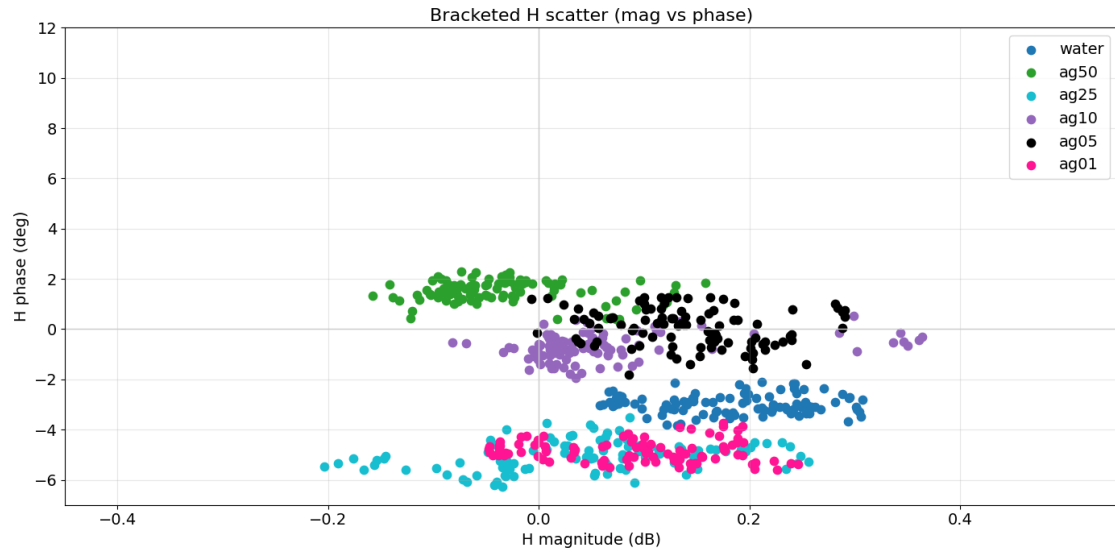


Figure 5.2: Scatterplot of bracketed magnitude versus phase for silver nanoparticles, for the transmission measurements.

Table 5.2: Transmission measurement summary statistics for silver at each concentration level.

Concentration (mg/mL)	Magnitude (dB)	Phase (deg)
0.00	0.186 ± 0.069	-2.981 ± 0.390
0.01	0.092 ± 0.081	-4.853 ± 0.432
0.05	0.142 ± 0.070	0.062 ± 0.748
0.10	0.065 ± 0.095	-0.700 ± 0.512
0.25	0.043 ± 0.106	-5.052 ± 0.579
0.50	-0.032 ± 0.068	1.494 ± 0.453

5.2 Reflection Measurements

The summary statistics for the reflection measurements of gold and silver nanoparticles are shown in Table 5.3 and Table 5.4 respectively.

Table 5.3: Reflection measurement summary statistics for gold at each concentration level.

Concentration (mg/mL)	Magnitude (dB)	Phase (deg)
0.00	0.105 ± 0.074	0.465 ± 0.785
0.01	0.057 ± 0.075	-0.566 ± 0.683
0.02	0.006 ± 0.068	8.636 ± 1.024
0.04	0.293 ± 0.079	-1.222 ± 0.913
0.05	-0.206 ± 0.058	-3.269 ± 0.748
0.08	0.070 ± 0.138	6.274 ± 0.589

Table 5.4: Reflection measurement summary statistics for silver at each concentration level.

Concentration (mg/mL)	Magnitude (dB)	Phase (deg)
0.00	0.105 ± 0.074	0.465 ± 0.785
0.01	0.106 ± 0.147	7.732 ± 1.045
0.05	-0.009 ± 0.139	1.497 ± 0.508
0.10	-0.049 ± 0.131	-2.400 ± 1.113
0.25	-0.200 ± 0.085	9.836 ± 0.812
0.50	0.092 ± 0.098	7.675 ± 0.838

Figure 5.3 and Figure 5.4 show the scatterplot of the standard sample bracketed magnitude vs phase for the reflection measurements of the gold and silver nanoparticles respectively.

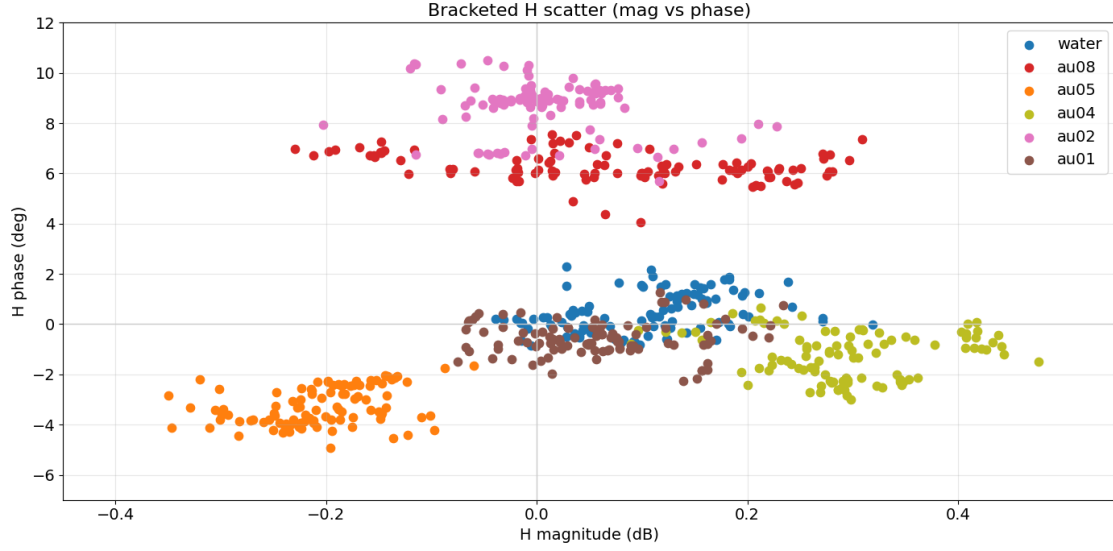


Figure 5.3: Scatterplot of bracketed magnitude versus phase for gold nanoparticles, for the reflection measurements.

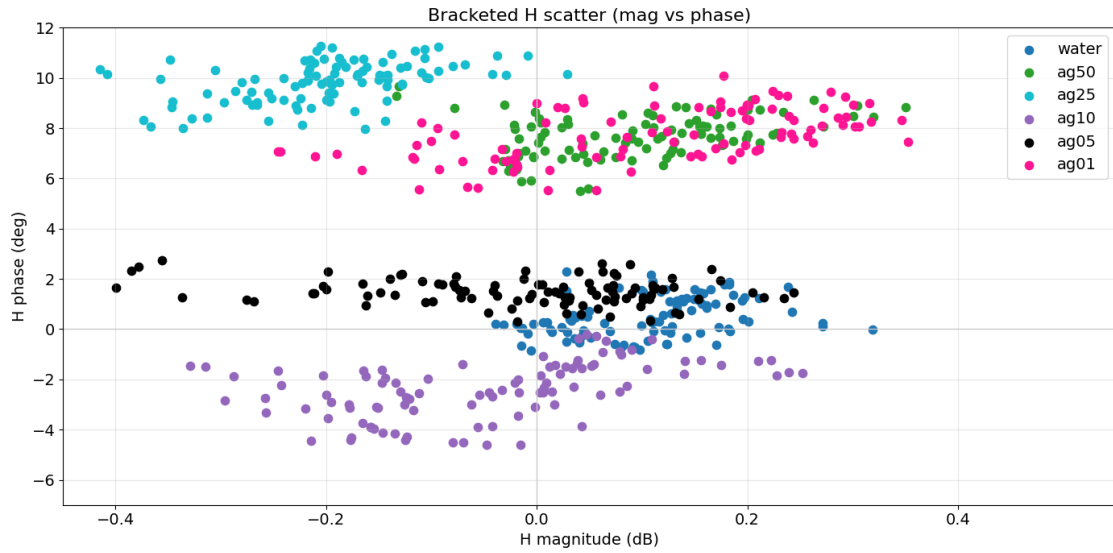


Figure 5.4: Scatterplot of bracketed magnitude versus phase for silver nanoparticles, for the reflection measurements.

5.3 Summary of Concentration-Dependent Trends

As the reported values are obtained from bracketed measurements, the focus is on the trend with concentration rather than the absolute measured values of the magnitude and phase. Across the four measurement cases, some variables showed a weak or moderate concentration-dependent tendency, while others appeared to be non-monotonic. In general, the phase responses often exhibited larger shifts across concentration levels, but these shifts were not always monotonic against concentration levels.

The measured magnitude and phase trends against concentration levels are visualized from Figure 5.5 to Figure 5.8.

For the transmission measurements of gold nanoparticles, neither the magnitude nor the phase showed a strong monotonic relationship with concentration. The transmission magnitude fluctuated across concentration levels, indicating no clear direct or inverse relationship. The transmission phase likewise did not follow a consistent monotonic pattern over the full concentration range, although after reaching a local peak at 0.02 mg/mL it generally decreased at the higher concentration levels. Overall, the gold transmission response might be described as weakly concentration-dependent but non-monotonic.

For the transmission measurements of silver nanoparticles, the magnitude showed a clear trend where as concentration increased, the transmission magnitude generally decreased, indicating a moderate to strong inverse relationship, despite some fluctuation at intermediate concentrations. In contrast, the transmission phase did not show a clear direct or inverse relationship with concentration, as the phase shifted irregularly between negative and positive values across the tested levels.

For the reflection measurements of gold nanoparticles, neither magnitude nor phase appeared to show a clear monotonic trend. The reflection magnitude rose and fell across the tested concentration levels, indicating no stable direct or inverse relationship. Similarly, the reflection phase alternated between positive and negative excursions, suggesting sensitivity to concentration changes, but not in a consistent monotonic direction. Hence, the gold nanoparticle's reflection response did not appear to show a clear direct or inverse relationship with concentration for both magnitude and phase.

For the reflection measurements of silver nanoparticles, the magnitude showed a weak inverse tendency up to 0.25 mg/mL, where the mean magnitude became progressively lower, but this pattern did not persist at 0.50 mg/mL due to a rebound in the mean value. As such, the overall relationship appeared to be weakly inverse and not fully monotonic. The reflection phase did not show very distinct trends, with large spikes to the phase at 0.01 and 0.25 mg/mL before exhibiting an inverse relationship as the concentration level increased. Overall, the silver nanoparticle's reflection phase may be described as concentration-sensitive but non-monotonic against concentration.

Taken together, the most evident concentration-dependent trend in the present results is the inverse relationship between concentration and transmission magnitude for silver nanoparticles. The remaining cases show either only weak tendencies or clearly non-monotonic behaviour. This suggests that, under the present experimental conditions, some measurement responses vary with concentration, but a simple linear or monotonic direct/inverse relationship is not uniformly observed across all gold and silver transmission and reflection measurements.

5.4 Comparison Against HFSS Simulations

The experimental results in section 5.3 can be compared against the expected concentration-dependent relationships predicted from the HFSS simulations in Table 3.1.

Overall, the HFSS simulations only partially matched the subsequently observed physical results. The simulations were helpful in identifying the 1.8 GHz region as a reasonable operating frequency, and the earlier sweep in section 3.3 had already shown that the measured transmission response peaked broadly in the same frequency region. The simulations had also indicated that the concentration-induced differences would be very subtle. However, the monotonic relationships predicted at 1.8 GHz were not reproduced consistently across the four measurement cases in section 5.3.

For the transmission measurements, the simulations had predicted a positive relationship between magnitude and concentration levels, and a negative relationship between phase and concentration. In the physical measurements, the transmission

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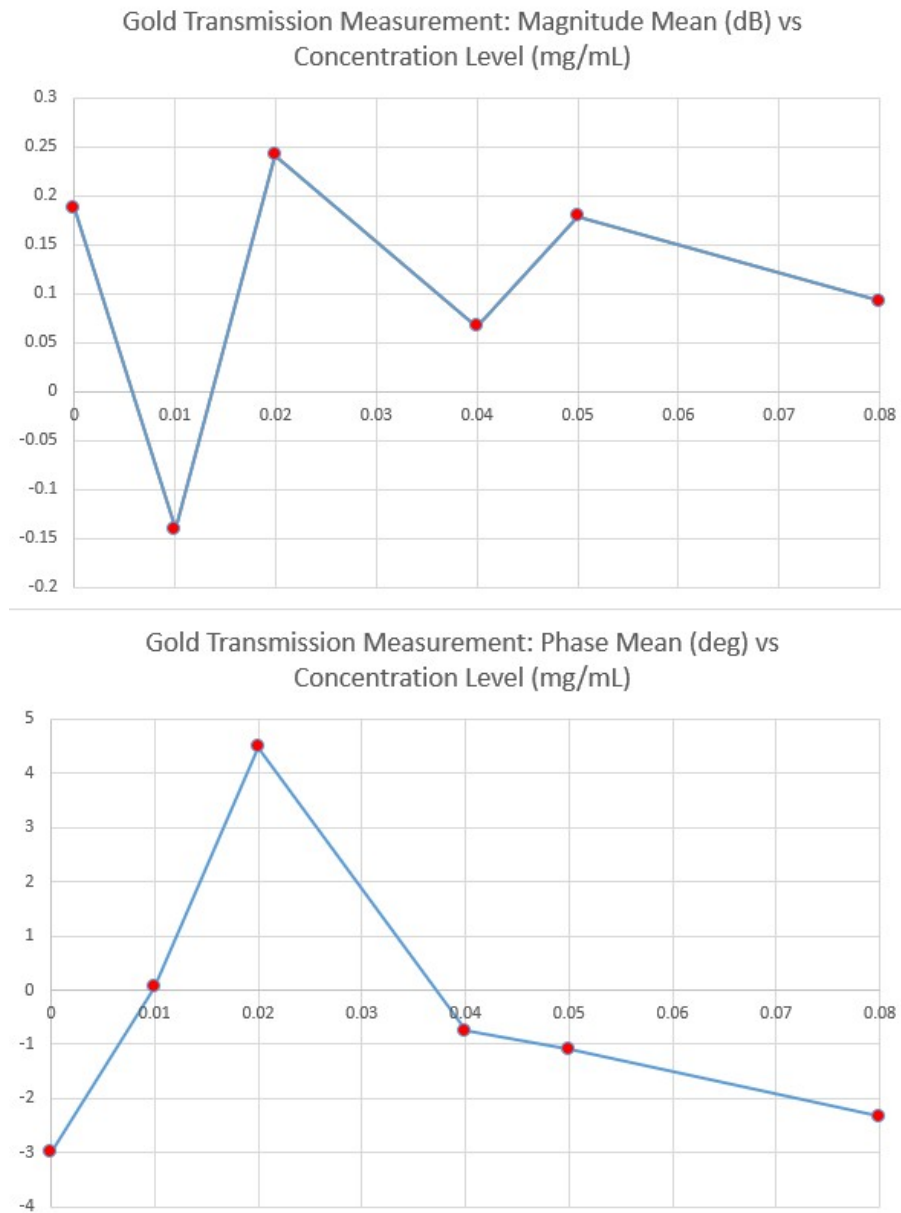


Figure 5.5: Transmission measurement trends for magnitude and phase vs concentration levels for gold samples.

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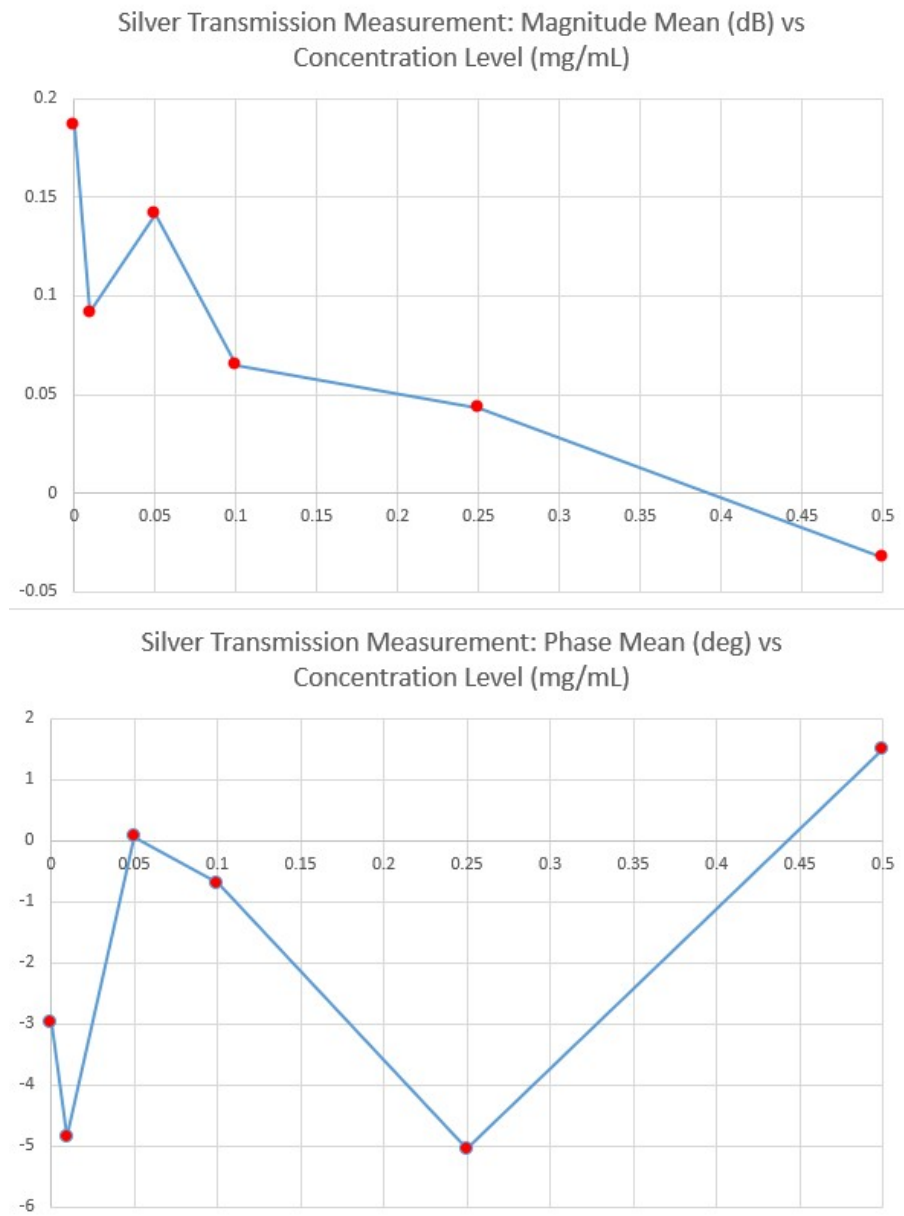


Figure 5.6: Transmission measurement trends for magnitude and phase vs concentration levels for silver samples.

CHAPTER 5. EXPERIMENTAL RESULTS

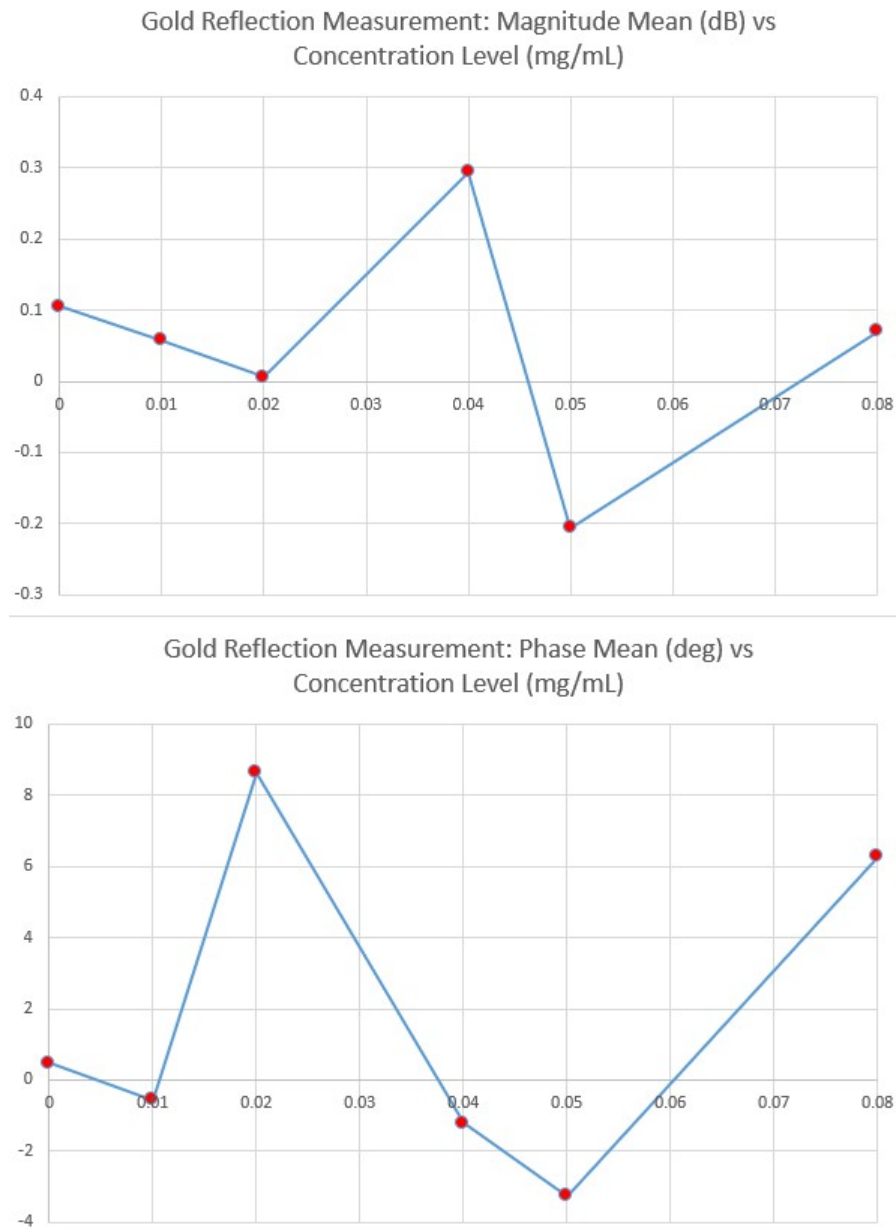


Figure 5.7: Reflection measurement trends for magnitude and phase vs concentration levels for gold samples.

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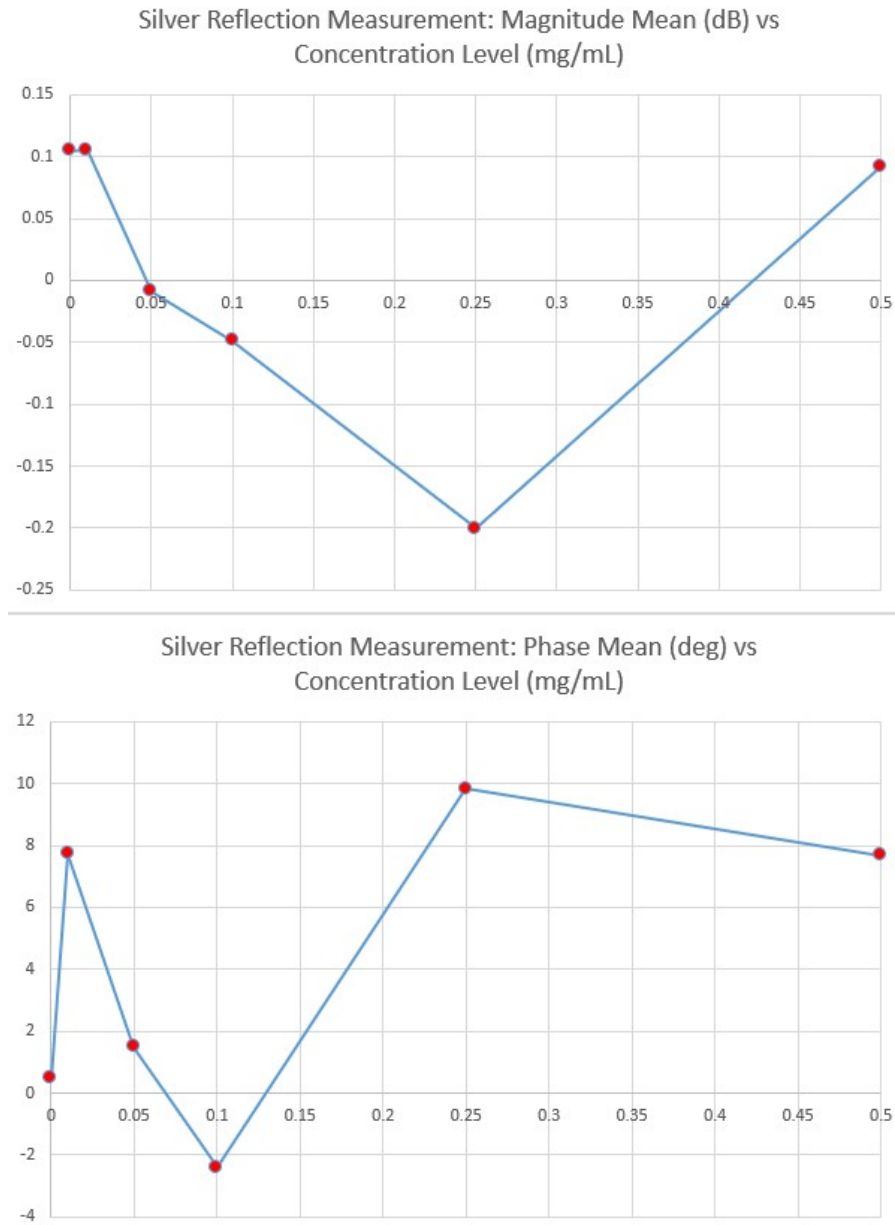


Figure 5.8: Reflection measurement trends for magnitude and phase vs concentration levels for silver samples.

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magnitude of the gold samples did not show a clear monotonic trend, while the silver samples instead showed a moderate inverse relationship. The transmission phase of the gold samples only showed a weak downward tendency at higher concentration levels, while the silver samples again did not exhibit a clear monotonic pattern. Hence, the transmission-phase result for gold may be regarded as only weakly consistent with simulation, whereas the other transmission results did not exactly match the simulated expectation.

It should be noted that Figure 3.4 suggested that 1.8 GHz lies relatively close to the transition from a direct to an inverse relationship between magnitude and concentration level. As such, the moderate inverse relationship observed for silver transmission magnitude cannot be discounted as being wholly inconsistent with the HFSS prediction. Instead, this may reflect the sensitivity of the response in this boundary region, where small shifts in the effective operating frequency or measurement conditions could plausibly alter the apparent trend direction.

For the reflection measurements, the simulations had predicted a negative relationship between magnitude and concentration levels, and a positive relationship between phase and concentration. Reflection measurements of the gold samples did not show a stable monotonic relationship for either magnitude or phase against concentration levels. The silver samples' reflection magnitude showed a weak inverse tendency up to intermediate concentration levels, which may be regarded as partial agreement with simulation, although this relationship broke down at the highest concentration level. The reflection phase results for both gold and silver were non-monotonic against concentration levels, and therefore did not provide strong support for the simulated positive relationship.

Taken together, the physical results did not appear to reproduce the simulated monotonic relationships reliably enough to claim direct validation of all four expected trends.

There are several plausible reasons for this discrepancy. First, the simulated shifts were extremely small in extent, so even modest experimental instability could obscure the expected monotonic behaviour, despite employing noise suppression using standard sample bracketing and selecting a frequency point that tries to maximize transmission response. Second, the selected operating frequency of 1.8 GHz lay close to transition regions in the HFSS results, especially for the transmis-

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sion and reflection magnitude responses, where the relationship with concentration changed sign nearby. This means that a small shift in the effective operating frequency caused by antenna mismatch, cable effects, manufacturing tolerances, or slight detuning of the real setup could possibly alter the observed trend direction.

Third, the HFSS model necessarily simplified the physical measurement geometry and sample behaviour. The simulation assumed an idealized and repeatable arrangement, whereas the real measurements were affected by practical factors such as slight variations in sample placement, liquid distribution within the well plate, meniscus shape and external electromagnetic interference. In particular, extended storage of the samples causing evaporation could also have affected the results, with the gold nanoparticle samples losing nearly half the original 1 mL volume through repeated pipetting in and out of the viewplates. These factors could have introduced additional phase and magnitude variation that was not captured in the idealized model.

Finally, the HFSS simulations were based on a simplified effective-medium representation using gold and water as the baseline material models. Although the gold and silver nanoparticle samples were expected to show broadly similar response at UHF frequencies, the physical samples may still differ in ways not fully reflected by the model, including dispersion and aggregation effects. As such, the simulations should be interpreted primarily as a guide for anticipating subtle concentration sensitivity, rather than as an exact predictor of every measured trend.

Chapter 6

Machine Learning Classification

6.1 Learning Task

After establishing that measurable differences in RF response exist across the tested samples, the next step was to try and determine whether these measured features are sufficiently consistent and informative for a machine learning model to distinguish concentration levels automatically. More specifically, the aim was to assess whether, given the type of nanoparticle solution (gold or silver), and given the type of measurement (transmission or reflection), the processed measurements can support practical concentration-level classification.

Accordingly, the problem was formulated as a supervised multiclass classification task. For each experimental setting, defined by the nanoparticle type (gold or silver) and the measurement mode (transmission or reflection), a separate classifier was trained to distinguish water, and the different nanoparticle concentration classes within that specific setting.

6.2 Random Forest Classifier

A random forest classifier is an ensemble learning method based on multiple decision trees. A decision tree is a sequence of binary splitting rules applied to the input features, such that each sample is progressively separated into smaller groups. The following discussion on how the random forest classifier works in general is based largely on the analysis by Louppe. [34]

At each internal node of a tree, the data is split according to a rule of the form: $x_j \leq t$, where x_j denotes the selected feature and t denotes a threshold value.

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Samples satisfying the rule are sent to one branch, while the remaining samples are sent to the other. Repeated application of such splits produces a tree-like structure, with each terminal node corresponding to a predicted class.

The split at each node is chosen so that samples belonging to the same class are grouped together as much as possible. A common measure of node impurity is the Gini impurity, given by

$$G = 1 - \sum_{k=1}^K p_k^2 \quad (6.1)$$

where p_k is the proportion of samples in the node belonging to class k , and K is the total number of classes. Lower impurity indicates that the node contains samples from fewer classes.

For a candidate split, the quality of the split may be assessed using the weighted impurity of the resulting child nodes, given by:

$$G_{\text{split}} = \frac{N_{\text{left}}}{N} G_{\text{left}} + \frac{N_{\text{right}}}{N} G_{\text{right}} \quad (6.2)$$

where N is the total number of samples in the parent node, while N_{left} and N_{right} denote the number of samples sent to the left and right child nodes respectively. The preferred split is the one that minimises this weighted impurity. [35]

Rather than relying on a single decision tree, a random forest constructs many trees using differing subsets of the training data and feature set. This introduces variation across the trees and reduces the tendency of the model to overfit to any single pattern in the training set.

For classification, each tree produces a predicted class, and the final output of the random forest is determined by majority voting across all trees. If the prediction of the b -th tree is denoted by $\hat{y}^{(b)}$, then the final predicted class $\hat{y}$ is expressed as:

$$\hat{y} = \text{mode}(\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(B)}), \quad (6.3)$$

where B is the total number of trees in the forest.

In this study, the random forest classifier was deemed suitable because it can model non-linear relationships and interactions between features without requiring heavy pre-processing, making it a practical choice for the learning task.

6.3 Feature Representation

In the dataset used for the random forest classifier, the required columns were `class`, `H_mag_k_db`, and `H_phase_k_deg`. Here, `class` is the target label, with possible classes being `water`, `au01`, `au02`, `au04`, `au05`, `au08`, `ag01`, `ag05`, `ag10`, `ag25` and `ag50`.¹ `H_mag_k_db` and `H_phase_k_deg` denote the processed magnitude and phase features derived from the bracketed RF response measurements.

The processed phase θ was first converted from degrees to radians and then represented using its sine and cosine components, to avoid discontinuities at the phase wrapping boundaries of $\pm 180^\circ$. Thus, each input sample was represented by:

$$\mathbf{x} = [|H_k|_{\text{dB}}, \sin \theta_k, \cos \theta_k] \quad (6.4)$$

where $|H_k|_{\text{dB}}$ denotes the magnitude in dB, and θ_k denotes the phase corresponding to the k -th processed measurement.

6.4 Model Setup and Evaluation

For each experiment setting (transmission for gold, transmission for silver, reflection for gold, reflection for silver), a random forest classifier was trained using the processed feature representation described earlier. For each model evaluation run, the data set was divided into training and test sets using an 80/20 split, with stratification by class so that the class proportions were preserved across both subsets. This procedure was repeated over 10 random seeds, in order to reduce dependence on any one particular train-test partition.

The random forest hyperparameters were configured with 200 trees, a maximum tree depth of 10, minimum sample split size of 5, minimum sample leaf size of 2, square-root feature sampling at each split, and no class-weight adjustment.

This configuration was informed by the earlier tuning process, in which various hyperparameter combinations were compared, and a setting that provided strong mean cross-validation accuracy on the training set while maintaining reasonable computational cost was selected. Here, mean cross-validation accuracy refers to the

¹`au` and `ag` denote gold and silver respectively, while the final two digits indicate the concentration level in units of 0.01 mg/mL. For example, `au01` denotes 0.01 mg/mL of gold concentration, and `ag50` denotes 0.50 mg/mL of silver concentration.

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average classification accuracy obtained across five internal train-validation partitions of the training data during the tuning process.

For each random seed, the data were partitioned into training and test sets using the same 80/20 stratified procedure, and the trained model was evaluated on the corresponding test set using classification accuracy and macro-averaged F_1 score. The mean and standard deviation of these test metrics were computed to summarize the overall performance and its variability across these random seeds.

The classification accuracy measures the proportion of correctly classified samples, while macro-averaged F_1 score is the average of the F_1 scores computed separately for each class, giving equal weight to all classes. The F_1 score itself is defined as the harmonic mean of precision and recall, and provides a single measure that balances both quantities:

$$F_1 = \frac{2PR}{P + R} \quad (6.5)$$

where precision and recall are defined respectively as: $P = \frac{TP}{TP+FP}$ and $R = \frac{TP}{TP+FN}$, with TP , FP , and FN denoting true positives, false positives, and false negatives.

One representative seed was selected for more detailed reporting, on the basis of having the test accuracy and test macro F_1 score jointly closest to their corresponding mean values across all repeated runs. For this representative seed, a full classification report and confusion matrix were generated and used for the detailed results discussion in section 6.5.

6.5 Results

The overall random forest classification performance for the four experimental settings is summarized in Table 6.1:

Table 6.1: Overall random forest classification performance across the four experimental settings, summarized over 10 repeated stratified 80/20 train-test runs.

Setting	Test accuracy	Macro F_1 score	Representative seed
Transmission (Gold)	0.8425 ± 0.0250	0.8435 ± 0.0243	1
Transmission (Silver)	0.8100 ± 0.0400	0.8095 ± 0.0399	4
Reflection (Gold)	0.8658 ± 0.0220	0.8662 ± 0.0224	0
Reflection (Silver)	0.8067 ± 0.0372	0.8040 ± 0.0372	9

Overall, these results indicate that the processed magnitude and phase features were sufficiently informative for multiclass classification in all four settings, although the degree of separability varied between them. The gold-based settings, particularly reflection-gold, exhibited the strongest and most stable performance across repeated random seeds, while the silver-based settings showed greater variability and somewhat weaker class separation.

6.5.1 Transmission Measurement (Gold)

For the transmission-gold setting, the representative run achieved a test accuracy of 0.8417 and a macro- F_1 score of 0.8405, as shown in Figure 6.1 and Table 6.2. The confusion matrix indicates generally strong class separation, with particularly strong performance for the **au02** class, which achieved perfect precision, recall, and F_1 score in the representative run. The **au01** class was also classified well, while **au04**, **au05**, and **au08** showed moderate but still reasonably strong performance.

Figure 6.1: Representative confusion matrix for the transmission measurement (gold) setting. Rows denote true classes and columns denote predicted classes.

True class	Predicted class					
	water	au01	au02	au04	au05	au08
water	14	0	0	0	2	4
au01	0	19	0	1	0	0
au02	0	0	20	0	0	0
au04	0	2	0	16	2	0
au05	1	0	0	3	15	1
au08	2	0	0	0	1	17

The main source of error in this setting involved the **water** class, which had a comparatively lower recall of 0.70. In the representative confusion matrix, several water samples were misclassified as **au08**, and smaller numbers of errors were also observed between **au04** and **au05**. This suggests that, under transmission measurements, the learned feature representation separated some concentration classes well, but the boundary between water and certain lower-contrast nanoparticle responses remained less distinct.

Table 6.2: Class-wise classification metrics for the representative transmission measurement (gold) run.

Class	Precision	Recall	F_1 score
water	0.82	0.70	0.76
au01	0.90	0.95	0.93
au02	1.00	1.00	1.00
au04	0.80	0.80	0.80
au05	0.75	0.75	0.75
au08	0.77	0.85	0.81
Macro average	0.84	0.84	0.84

6.5.2 Transmission Measurement (Silver)

For the transmission-silver setting, the representative run achieved a test accuracy of 0.8083 and a macro- F_1 score of 0.8089, as shown in Figure 6.2 and Table 6.3. In this case, the **water** class was separated particularly clearly, achieving perfect precision and recall in the representative run. The **ag50** class also performed strongly, while the intermediate concentration classes showed more modest separation.

Figure 6.2: Representative confusion matrix for the transmission measurement (silver) setting. Rows denote true classes and columns denote predicted classes.

True class	Predicted class					
	water	ag01	ag05	ag10	ag25	ag50
water	20	0	0	0	0	0
ag01	0	12	0	0	8	0
ag05	0	0	15	5	0	0
ag10	0	0	4	16	0	0
ag25	0	4	0	0	16	0
ag50	0	0	2	0	0	18

The confusion matrix indicates that the principal errors arose between neighbouring silver concentration classes. In particular, notable confusion occurred between **ag01** and **ag25**, and between **ag05** and **ag10**. The **ag01** class had the weakest recall in the representative run, at 0.60, while **ag10** and **ag25** performed somewhat better. Overall, these results suggest that the transmission-silver setting supported meaningful classification, but that the class boundaries between several silver concentration levels were less clearly separated than in the case of transmission measurements for gold.

Table 6.3: Class-wise classification metrics for the representative transmission measurement (silver) run.

Class	Precision	Recall	F_1 score
water	1.00	1.00	1.00
ag01	0.75	0.60	0.67
ag05	0.71	0.75	0.73
ag10	0.76	0.80	0.78
ag25	0.67	0.80	0.73
ag50	1.00	0.90	0.95
Macro average	0.82	0.81	0.81

6.5.3 Reflection Measurement (Gold)

For the reflection-gold setting, the representative run achieved a test accuracy of 0.8667 and a macro- F_1 score of 0.8654, as shown in Figure 6.3 and Table 6.4. This was the strongest-performing setting overall. The representative class-wise results show that **au05** was classified perfectly, while **au02**, **au04**, and **au08** also achieved high precision, recall, and F_1 scores. These results indicate that, for reflection measurements of gold nanoparticles, the processed features gave comparatively strong discrimination across multiple concentration levels.

Figure 6.3: Representative confusion matrix for the reflection measurement (gold) setting. Rows denote true classes and columns denote predicted classes.

True class	Predicted class					
	water	au01	au02	au04	au05	au08
water	15	4	0	1	0	0
au01	6	13	0	1	0	0
au02	0	0	18	0	0	2
au04	0	1	0	19	0	0
au05	0	0	0	0	20	0
au08	0	0	1	0	0	19

The main remaining weakness was in the separation between **water** and **au01**. In the representative confusion matrix, several water samples were predicted as **au01**, and several **au01** samples were predicted as water. Accordingly, these two classes had the lowest class-wise F_1 scores in this setting. Nevertheless, the reflection-gold responses exhibited the clearest overall class structure among the four experimental settings considered.

Table 6.4: Class-wise classification metrics for the representative reflection measurement (gold) run.

Class	Precision	Recall	F_1 score
water	0.71	0.75	0.73
au01	0.72	0.65	0.68
au02	0.95	0.90	0.92
au04	0.90	0.95	0.93
au05	1.00	1.00	1.00
au08	0.90	0.95	0.93
Macro average	0.87	0.87	0.87

6.5.4 Reflection Measurement (Silver)

For the reflection-silver setting, the representative run achieved a test accuracy of 0.8083 and a macro- F_1 score of 0.8033, as shown in Figure 6.4 and Table 6.5. Although the overall performance was comparable to that of transmission-silver, the class-wise behaviour differed somewhat. In the representative run, **ag25** achieved perfect recall, and **ag10** also performed strongly, while **water**, **ag01**, and **ag50** were comparatively weaker.

Figure 6.4: Representative confusion matrix for the reflection measurement (silver) setting. Rows denote true classes and columns denote predicted classes.

True class	Predicted class					
	water	ag01	ag05	ag10	ag25	ag50
water	13	0	6	1	0	0
ag01	0	13	0	0	3	4
ag05	1	0	19	0	0	0
ag10	2	0	0	18	0	0
ag25	0	0	0	0	20	0
ag50	0	5	0	0	1	14

The confusion matrix shows that the main errors occurred between **water** and **ag05**, and between **ag50** and **ag01**. The **ag05**, **ag10**, and **ag25** classes were generally more separable, whereas the lower and higher ends of the class range exhibited more overlap in the learned feature space. Taken together, these results indicate that the reflection-silver setting still supported useful concentration-level classification, but with greater ambiguity for certain classes.

Table 6.5: Class-wise classification metrics for the representative reflection measurement (silver) run.

Class	Precision	Recall	F_1 score
water	0.81	0.65	0.72
ag01	0.72	0.65	0.68
ag05	0.76	0.95	0.84
ag10	0.95	0.90	0.92
ag25	0.83	1.00	0.91
ag50	0.78	0.70	0.74
Macro average	0.81	0.81	0.80

6.5.5 Overall Discussion

In summary, the random forest classifier achieved reasonably strong multiclass performance across all four settings, demonstrating that the processed RF response features contain class-discriminative information. However, the results also show that separability was not uniform across all materials, measurement modes, and concentration levels.

In particular, reflection-gold provided the strongest overall classification performance, while the silver-based settings exhibited more frequent overlap between some concentration classes. These observations are consistent with the earlier descriptive analysis, in which some experimental settings displayed clearer concentration-related trends than others.

Chapter 7

Conclusion

This study investigated whether UHF radio sensing could detect concentration-dependent differences in aqueous gold and silver nanoparticle solutions, as an initial exploratory step toward non-invasive sensing applications. HFSS simulation, controlled physical measurements, and downstream machine-learning analysis were combined to study the problem. Overall, the work showed that measurable RF magnitude and phase differences could be extracted from the nanoparticle samples under controlled conditions, although the concentration-dependent relationships were not uniformly monotonic across all the different experimental settings.

The experimental findings showed that the clearest descriptive trend was the inverse relationship between concentration and transmission magnitude for the silver samples, while the remaining measurement cases displayed either weaker tendencies or non-monotonic behaviour, as summarized in section 5.3. Comparison against the HFSS simulations showed only partial agreement. In particular, the simulations were still useful in identifying the 1.8 GHz region as a reasonable operating point and in indicating that the expected shifts would be subtle, but the predicted monotonic relationships at 1.8 GHz were not reproduced consistently in the physical measurements. Taken together, this suggests that while the simulations provided useful directional guidance, the real measurement response remained sensitive to practical experimental factors and to the precise operating frequency.

Despite this, the processed RF features still supported reasonably strong multiclass classification performance across all four settings, with mean test accuracies ranging from 0.8067 to 0.8658, and with reflection measurements of gold samples providing the strongest overall results, as shown in section 6.5. This indicates that even where the descriptive trends were not cleanly monotonic, the measure-

CHAPTER 7. CONCLUSION

ment pipeline was still able to extract class-discriminative information from the RF responses. As such, the main contribution of this study is to demonstrate the feasibility of combining UHF sensing, robust signal processing, and data-driven analysis to distinguish nanoparticle concentration levels in aqueous samples.

Future work may improve the measurement sensitivity further in several ways. First, because the results suggested that 1.8 GHz lay relatively close to transition regions in the simulated response, and because the initial exploratory frequency sweep also showed a broader peak in the surrounding frequency region, it might be worthwhile to explore nearby frequencies within the UHF band rather than fixing the study only at 1.8 GHz. A small shift to the left or right of 1.8 GHz may potentially produce more stable or more clearly monotonic concentration-dependent behaviour, thereby supporting more consistent downstream classification.

Second, future studies may also systematically investigate placing the sample closer to the antenna system, particularly as it approaches the near-field boundary, to determine whether stronger coupling can improve sensitivity to the sample-induced RF response. Such an approach would need to be balanced carefully against the greater geometry sensitivity and placement instability that motivated the present study to keep the sample outside the near-field region. More broadly, future work should also seek improved experimental stability, refined physical modelling, and validation on more biologically representative samples.

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